

Syllabus Topic : Types of IIR Filters (Low Pass, High Pass, Band Pass, Band Stop and All Pass)

2.1 IDEAL FILTER CHARACTERISTICS

GQ. Write short note on : Filter classification based on passband. (5 Marks)

Some parameters related to filters

1. Pass band
2. Stop band
3. Cut - off frequency

1. Pass band

- It passes certain range of frequencies. In the pass band, attenuation is zero.

2. Stop band

It suppresses certain range of frequencies. In the stop band, attenuation is infinity.

3. Cut-off frequency

This is the frequency which separates pass band and stop band.

Filter can be classified as follows

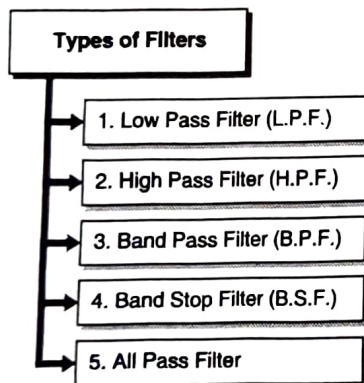


Fig. 2.1.1 : Types of filters

1. Low Pass Filter (L.P.F.)

- It passes the frequency from zero upto some designated frequency, called as cut-off frequency.
- After cut-off frequency; it will not allow any signal to pass through it. An ideal low pass filter characteristic is shown in Fig. 2.1.2(a).

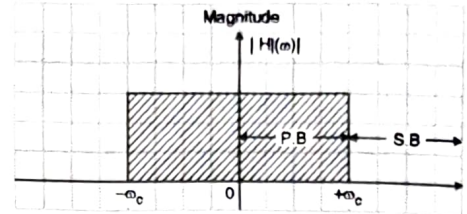


Fig. 2.1.2(a) : Ideal L.P.F. characteristic

2. High Pass Filter (H.P.F.)

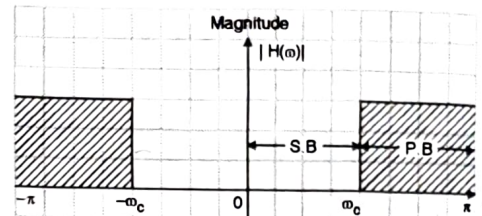


Fig. 2.1.2(b) : Ideal H.P.F. characteristic

- It passes the frequency above some designated frequency called as cut-off frequency.
- If input signal frequency is less than the cut-off frequency; then this signal is not allowed to pass through it. An ideal H.P.F. characteristic is shown in Fig. 2.1.2(b).

3. Band Pass Filter (B.P.F.)

It allows the frequencies between two designated cut-off frequencies (say f_{c1} and f_{c2}). An ideal B.P.F. characteristic is shown in Fig. 2.1.2(c).

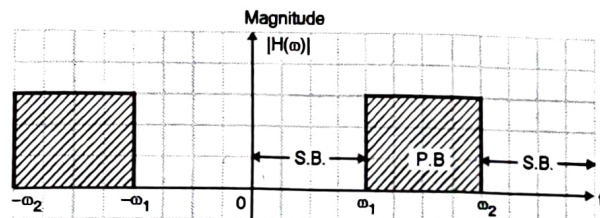


Fig. 2.1.2(c) : Ideal B.P.F. characteristic

4. Band Stop Filter (B.S.F.)

It attenuates all frequencies between two designated cut-off frequencies; while it passes all other frequencies. An ideal B.S.F. characteristic is shown in Fig. 2.1.2(d).

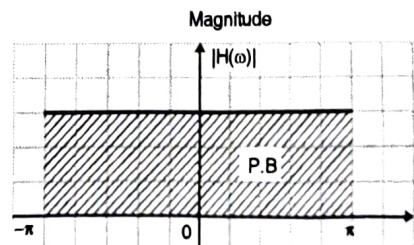


Fig. 2.1.2(d) : Ideal B.S.F. characteristic

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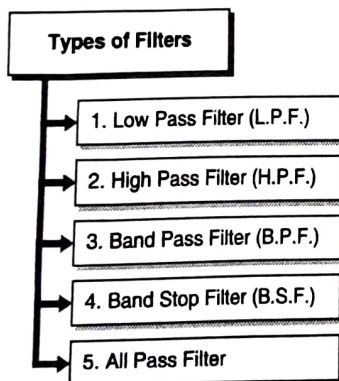


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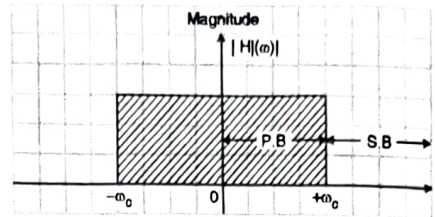


Fig. 2.1.2(a) : Ideal L.P.F. characteristic

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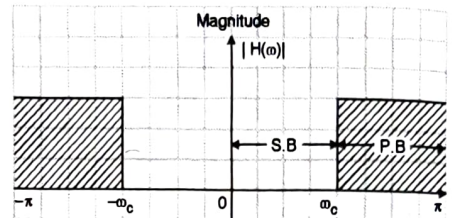


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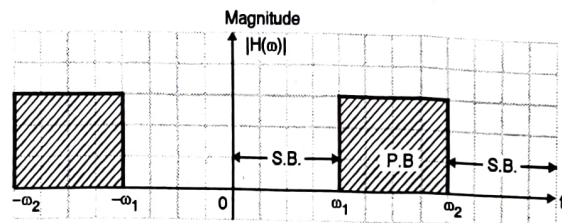


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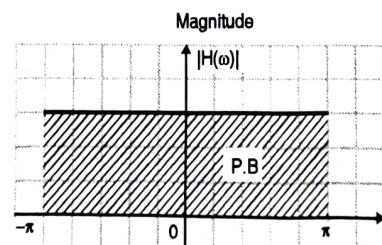


Fig. 2.1.2(d) : Ideal B.S.F. characteristic

5. All Pass Filter

It passes all the frequencies. But by using this filter the phase of input signal can be modified. All pass filter characteristic is shown in Fig. 2.1.2(e).

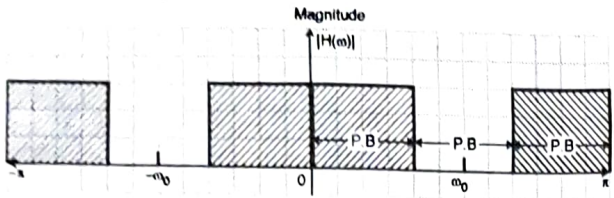


Fig. 2.1.2(e) : All pass filter characteristic

- Some important characteristics of ideal filter are as follows :
 - (i) Ideal filter has a linear phase response.
 - (ii) Ideal filter has a constant magnitude characteristic.
 - (iii) Ideal filter has a constant group delay.
- A group delay is a time delay that the signal components take when they are passing through the filter.
- It is denoted by $T_g(\omega)$ and it is derivative of phase with respect to frequency.

$$\therefore T_g(\omega) = \frac{d[\angle H(\omega)]}{d\omega}$$

2.1.1 Advantages of Digital Filters

UQ. State advantages of digital filters.

MU - Q. 1(e), May 2019, 5 Marks

1. Many input signals can be filtered by one digital filter without replacing the hardware.
2. Digital filters have characteristic like linear phase response. Such characteristic is not possible to obtain in case of analog filters.
3. The performance of digital filters, does not vary with environmental parameters. But the environmental parameters like temperature, humidity etc., change the values of components in case of analog filters. So it is required to calibrate analog filters periodically.
4. In case of digital filters; since the filtering is done with the help of digital computer, both filtered and unfiltered data can be saved for further use.
5. Unlike analog filters; the digital filters are portable.
6. From unit to unit the performance of digital filters is repeatable.
7. The digital filters are highly flexible.
8. Using VLSI technology; the hardware of digital filters can be reduced. Similarly the power consumption can be reduced.
9. Digital filters can be used at very low frequencies, for example in Biomedical applications.
10. In case of analog filters; maintenance is frequently required. But for digital filters it is not required.

2.1.2 Disadvantages of Digital Filters

1. Speed limitation

In case of digital filters, ADC and DAC are used. So the speed of digital filter depends on the conversion time of ADC and the settling time of DAC. Similarly the speed of operation of digital filter depends on the speed of digital processor. Thus the bandwidth of input signal processed is limited by ADC and DAC. In real time applications, the bandwidth of digital filter is much lower than analog filters.

2. Finite wordlength effect

The accuracy of digital filter depends on the wordlength used to encode them in binary form. Wordlength should be long enough to obtain the required accuracy.

The digital filters are also affected by the ADC noise, resulting from the quantization of continuous signals. Similarly the accuracy of digital filters is also affected by the roundoff noise occurred during computation.

3. Long design and development time

An initial design and development time for digital hardware is more than analog filters.

Syllabus Topic : Analog Filter Approximations : Butterworth, Chebyshev I, Elliptic

2.2 CHARACTERISTICS OF BUTTERWORTH AND CHEBYSHEV FILTERS

Many times it is necessary to approximate the characteristics of analog filter. This approximation is required because the practical characteristic of a filter is not identical to the ideal characteristics. There are three different types of approximation techniques as follows :

- (1) Butterworth filter approximation.
- (2) Chebyshev filter approximation.

► (1) Butterworth filter approximation

GQ. Write short note on butterworth filter approximation (5 Marks)

- A typical characteristic of a butterworth low pass filter is as shown in Fig. 2.2.1.
- This type of response is called as butterworth response because its main characteristic is that the passband is maximally flat. That means there are no variations (ripples) in the passband.

Now the magnitude squared response of low pass butterworth filter is given by,

$$|H(\Omega)|^2 = \frac{1}{1 + \left(\frac{\Omega}{\Omega_c}\right)^{2N}} \quad \dots(2.2.1)$$

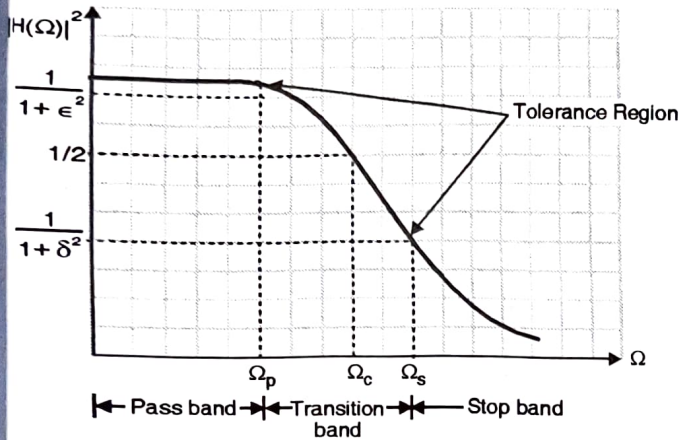


Fig. 2.2.1 : Typical characteristics of analog L.P.F.

This equation is also expressed as,

$$|H(\Omega)|^2 = \frac{1}{1 + \epsilon^2 \left(\frac{\Omega}{\Omega_p}\right)^{2N}} \quad \dots(2.2.2)$$

Here $|H(\Omega)|$ = Magnitude of analog low pass filter.

Ω_c = Cut-off frequency (- 3B frequency).

Ω_p = Pass band edge frequency.

$1 + \epsilon^2$ = Pass band edge value.

$1 + \delta^2$ = Stop band edge value.

ϵ = Parameter related to ripples in pass band.

δ = Parameter related to ripples in stop band.

N = Order of the filter.

We know that, in case of low pass filter the frequencies will pass upto the value of cut-off frequency (Ω_c). This is called as pass band.

After that the frequencies are attenuated. This is called as stop band. Ideal characteristic is shown by dotted line in Fig. 2.2.2.

Ideally, at the value of cut-off frequency (Ω_c) the frequencies should be stopped. But in practical cases this is not happening.

Now the order of filter is denoted by 'N'. Roughly we can say order of filter means, the number of stages used in the design of analog filter.

As the order of filter 'N' increase's, the response of filter is more close to the ideal response as shown in Fig. 2.2.2.

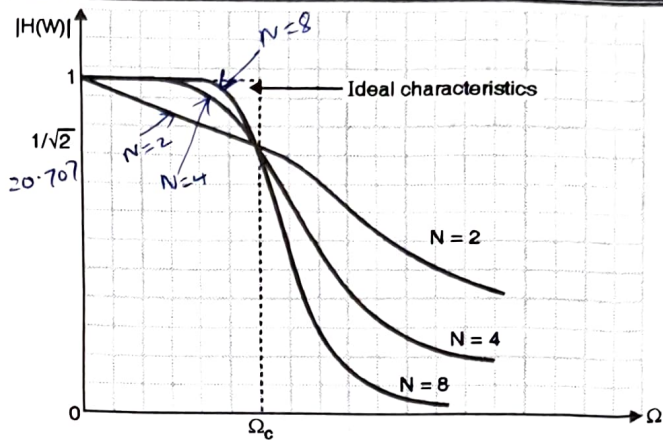


Fig. 2.2.2 : Effect of N on frequency response characteristics



Salient features of low pass butterworth filter

1. The magnitude response is nearly constant (equal to 1) at lower frequencies. That means pass band is maximally flat.
2. There are no ripples in the pass band and stop band.
3. The maximum gain occurs at $\Omega = 0$ and it is $|H(0)| = 1$.
4. The magnitude response is monotonically decreasing.

Table 2.2.1 : List of butterworth polynomials

N	Denominator of H(s)
1	$s + 1$
2	$s^2 + \sqrt{2}s + 1$
3	$(s + 1)(s^2 + s + 1)$
4	$(s^2 + 0.76537s + 1)(s^2 + 1.8477s + 1)$
5	$(s + 1)(s^2 + 0.61803s + 1)(s^2 + 1.61803s + 1)$
6	$(s^2 + 1.931855s + 1)(s^2 + \sqrt{2}s + 1)(s^2 + 0.51764s + 1)$
7	$(s + 1)(s^2 + 1.80194s + 1)(s^2 + 1.247s + 1)(s^2 + 0.445s + 1)$

(2) Chebyshev filter

Using chebyshev filter design, there are two subgroups,

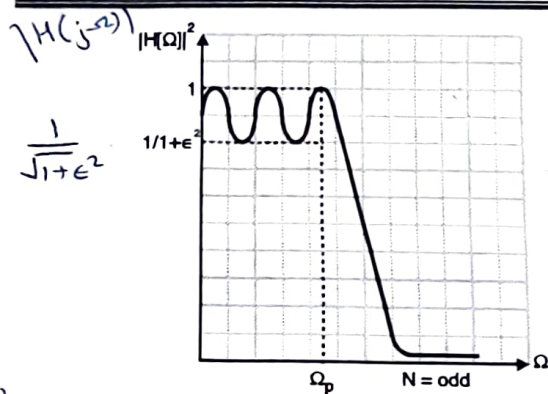
1. Type-1 chebyshev filter
2. Type-2 chebyshev filter

1. Type-1 Chebyshev Filter

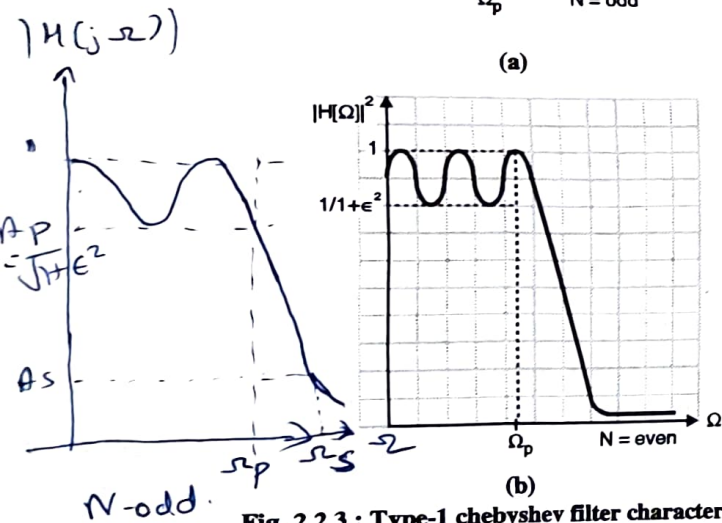
These filters are all pole filters. In the passband, these filters show equiripple behaviour and they have monotonic characteristics in the stopband.

The filter characteristics for odd and even values of N are shown in Fig. 2.2.3.

Refer Pg 7.44



(a)



(b)

Fig. 2.2.3 : Type-I chebyshev filter characteristics

- The magnitude squared frequency response is given by,

$$|H(\Omega)|^2 = \frac{1}{1 + \epsilon^2 C_N^2(\Omega/\Omega_p)} \quad \dots(2.2.3)$$

Here, ϵ = Ripple parameter in the passband ;

Ω_p = Passband frequency

$C_N(x)$ = Chebyshev polynomial of order N.

The chebyshev polynomials are determined by using the equation,

$$C_{N+1}(x) = 2x C_N(x) - C_{N-1}(x) \quad \dots(2.2.4)$$

with $C_0(x) = 1$ and $C_1(x) = x$

- The different polynomials are given in Table 2.2.2.

Table 2.2.2

N	$C_N(x)$
0	1
1	x
2	$2x^2 - 1$
3	$4x^3 - 3x$
4	$8x^4 - 8x^2 + 1$
5	$16x^5 - 20x^3 + 5x$

- The major difference between butterworth and chebyshev filter is that the poles of butterworth filter lie on the circle, while the poles of chebyshev filter lie on ellipse.

2. Type-II Chebyshev Filter

- This filter contains zeros as well as poles. The characteristic of such filters for even and odd values of N is shown in Fig. 2.2.4.

- The magnitude squared response is given by,

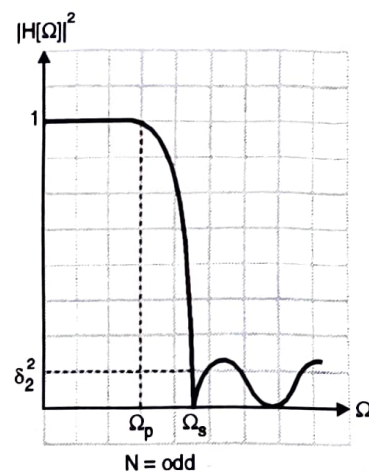
$$|H(\Omega)|^2 = \frac{1}{1 + \epsilon^2 \left[C_N^2\left(\frac{\Omega_s}{\Omega}\right) / C_N^2\left(\frac{\Omega}{\Omega_p}\right) \right]}$$

Here,

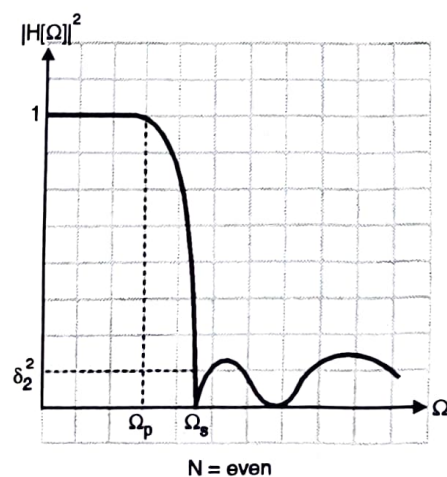
$C_N(x)$ = N^{th} order polynomial;

Ω_s = Stopband frequency;

Ω_p = Passband frequency



(a)



(b)

Fig. 2.2.4 : Type-II chebyshev filter characteristic

2.3 ELLIPTIC FILTERS

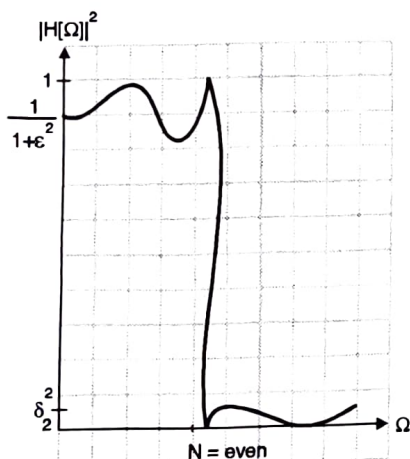
- Elliptic filters are also called as cauer filters. Such filters have equiripple passband and stopband.
- The magnitude squared frequency response of elliptic filter is shown in Fig. 2.3.1.

$$|H(\Omega)|^2 = \frac{1}{1 + \epsilon^2 U_N^2\left(\frac{\Omega}{\Omega_p}\right)} \quad \dots(2.3.1)$$

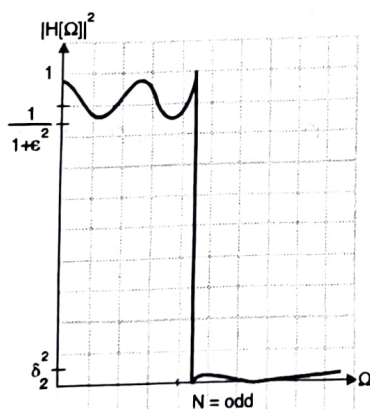
Here, ϵ = Ripple parameter in passband

Ω_p = Passband edge frequency

$U_N(x)$ = Jacobian elliptic function of order N.



(a)



(b)

Fig. 2.3.1 : Elliptic filter characteristics

The order 'N' of elliptic filter is given by,

$$N = \frac{k(\Omega_p / \Omega_s) k(\sqrt{1 - (\epsilon^2 / \delta^2)})}{k(\epsilon / \delta) k(\sqrt{1 - (\Omega_p / \Omega_s)^2})}$$

Here δ is related to δ_1 and δ_2 as follows :

$$\delta_2 = \text{stopband ripple}$$

$$\text{and } \delta_1 = 10 \log_{10}(1 + \epsilon^2)$$

where δ_1 = stopband ripple

The function $k(x)$ is called as elliptic integral of first kind.

Advantages of elliptic filters

1. Elliptic filter is more efficient because it provides the smallest order filter for given set of specifications.
2. It has smallest transition bandwidth.

Disadvantage of elliptic filters

The phase response is more non-linear in passband.

Syllabus Topic : Design of IIR Digital Filters (Butterworth and Chebyshev-I with Examples)

2.4 DESIGN OF IIR DIGITAL FILTERS (BUTTERWORTH AND CHEBYSHEV WITH EXAMPLES)

- The basic function of digital filter is to eliminate the noise and to extract the signal of interest from other signals. A digital filter is a basic device used in digital signal processing.
- There are several techniques available to design the digital filters. But generally while designing a digital filter; first an analog filter is designed and then it is converted into the corresponding digital filter.
- In order to design the digital IIR filter; analog IIR filter is designed first. Then analog filter is converted into the digital filter. Here you may ask a question, why to design digital filter from analog filter ?
- The reasons are as follows :
 1. The procedure to design analog filter is readily available and it is highly advanced.
 2. When we design digital filter using analog filter then the implementation becomes simple.

The different methods to design IIR filters are :

1. Impulse invariance
2. Bilinear transformation.
3. Matched Z transformation

**Syllabus Topic : Mapping of S-plane to Z-plane
Impulse Invariance Method**

2.5 IMPULSE INVARIANT METHOD

UQ. Show the mapping from S-plane to Z plane using impulse invariance method and explain its limitation.

MU - May 05, May 06, May 16, 3 Marks

UQ. Derive the expression for impulse invariance technique for obtaining transfer function of digital filter from analog filter. Derive the necessary equation for relationship between frequency of analog and digital filter. **MU - Dec. 11, 8 Marks**

UQ. Explain the mapping from S-plane to Z-plane using impulse invariance technique. Also explain the limitations of this method.

MU - Q. 3(b), Dec. 18, 10 Marks

- In this method, the design starts from the specifications of analog filter. Here we have to replace analog filter by digital filter.
- This is achieved if impulse response of digital filter resembles the sampled version of impulse response of analog filter. If impulse response of both, analog and digital filter matches then, both filters perform in a similar manner.
- Before studying this method we will list out the different notations, we are going to use.

$h(t)$ = Impulse response in time domain (analog domain)

$H_a(s)$ = Transfer function of analog filter; here 's' is Laplace operator

$h(nT_s)$ = Sampled version of $h(t)$, obtained by replacing t by nT_s .

$H(Z)$ = Z transform of $h(nT_s)$. This is response of digital filter. Or Transfer function of the filter in digital domain.

Ω = Analog frequency

ω = Digital frequency

Transformation of analog system function $H_a(s)$ to digital system function $H(Z)$

- Now let the system transfer function of analog filter be $H_a(s)$. We can express $H_a(s)$ in terms of partial fraction expansion. That means,

$$H_a(s) = \frac{A_1}{s-P_1} + \frac{A_2}{s-P_2} + \frac{A_3}{s-P_3} \dots \quad (*)$$

$$\therefore H_a(s) = \sum_{k=1}^N \frac{A_k}{s-P_k} \quad \dots(2.5.1)$$

Here, $A_k = A_1, A_2 \dots A_N$ are the coefficients of partial fraction expansion.

and $P_k = P_1, P_2 \dots P_N$ are the poles.

- Here 's' is the laplace operator. So we can obtain impulse response of analog filter, $h(t)$ from $H_a(s)$ by taking inverse laplace of $H_a(s)$. So using standard relation of inverse laplace we get.

$$h(t) = \sum_{k=1}^N A_k e^{P_k t} \quad \dots(2.5.2)$$

- Now unit impulse response for discrete structure is obtained by sampling $h(t)$. That means, $h(n)$ can be obtained from $h(t)$ by replacing 't' by nT_s in Equation (2.5.2). $t = nT_s$

$$\therefore h(n) = \sum_{k=1}^N A_k e^{P_k nT_s} \quad \dots(2.5.3)$$

Here T_s is the sampling time.

- The system transfer function of digital filter is denoted by $H(Z)$. It is obtained by taking Z - transform of $h(n)$. According to the definition of Z - transform for causal system,

$$H(Z) = \sum_{n=0}^{\infty} h(n) Z^{-n} \quad \dots(2.5.4)$$

- Putting Equation (2.5.3) in Equation (2.5.4) we get,

$$H(Z) = \sum_{n=0}^{\infty} \left[\sum_{k=1}^N A_k e^{P_k nT_s} \right] \cdot Z^{-n}$$

$$\therefore H(Z) = \sum_{k=1}^N A_k \sum_{n=0}^{\infty} e^{P_k T_s n} \cdot Z^{-n}$$

$$\therefore H(Z) = \sum_{k=1}^N A_k \sum_{n=0}^{\infty} (e^{P_k T_s} \cdot Z^{-1})^n \quad \dots(2.5.5)$$

- Using the standard summation formula,

$$\sum_{n=0}^{\infty} a^n = \frac{1}{1-a}, \quad |a| < 1$$

- Equation (2.5.5) becomes,

$$H(Z) = \sum_{k=1}^N A_k \cdot \frac{1}{1 - e^{P_k T_s} \cdot Z^{-1}} \quad \dots(2.5.6)$$

- This is the required transfer function of digital filter.

- Thus comparing Equations (2.5.1) and (2.5.6), we can say that the transfer function of digital filter is obtained from the transfer function of analog filter by doing the transformation.

$$\frac{1}{s-P_k} \longrightarrow \frac{1}{1 - e^{P_k T_s} \cdot Z^{-1}} \quad \dots(2.5.7)$$

- Equation (2.5.7) shows, how the poles from analog domain are transferred into the digital domain. This transformation of poles is called as **mapping of poles**.

This transformation is called Impulse Invariance mapping of poles.

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Computing the ILT of eqn
 $L^{-1} \left[\frac{1}{s-a} \right] = e^{at} u(t)$
 $f(s) = \frac{A_1}{s-P_1} + \frac{A_2}{s-P_2} + \frac{A_3}{s-P_3} + \dots + \frac{A_N}{s-P_N}$

Relationship of s-plane to Z plane

We know that the poles of analog filters are located at $s = P_k$. Now from Equation (2.5.7) we can say that the poles of digital filter, $H(Z)$ are located at,

$$Z = e^{P_k T_s} \quad \dots(2.5.8)$$

- This equation indicates that the poles of analog filter at $s = P_k$ are transformed into the poles of digital filter at $Z = e^{P_k T_s}$.
- Thus the relationship between laplace ('s' domain) and Z domain is given by,

$$Z = e^{s T_s} \quad \dots(2.5.9)$$

Here $s = P_k$ and T_s is the sampling time.

- Now 's' is the laplace operator and it is expressed as,

$$s = \sigma + j\Omega \quad \dots(2.5.10)$$

Here σ = Attenuation factor

and Ω = Analog frequency

- We know the 'Z' can be expressed in polar form as,

$$Z = r e^{j\omega} \quad \dots(2.5.11)$$

Here 'r' is magnitude and ' ω ' is the digital frequency.

- Putting Equations (2.5.10) and (2.5.11) in Equation (2.5.9) we get,

$$r e^{j\omega} = e^{(\sigma + j\Omega) T_s} \quad \dots(2.5.12)$$

- Separating real and imaginary parts of Equation (2.5.12) we get,

$$r = e^{\sigma T_s} \quad \dots(2.5.13)$$

$$\text{and } e^{j\omega} = e^{j\Omega T_s} \quad \dots(2.5.14)$$

- Now we will find the relationship between s plane and Z plane. Basically plot in 's'-domain means, σ is plotted on X-axis and $j\Omega$ is plotted on Y-axis. And Z-domain representation means real Z is plotted on X-axis and imaginary Z is plotted on Y-axis.

- Now consider Equation (2.5.13), it is

$$r = e^{\sigma T_s}$$

We will discuss the following conditions :

- (i) If $\sigma < 0$, then r is equal to reciprocal of 'e' raise to some constant. Thus range of r will be 0 to 1.

$$\therefore \sigma < 0 \Rightarrow 0 < r < 1 \quad \dots(2.5.15)$$

Now $\sigma < 0$ means negative values of σ . That is L.H.S. of s plane. We know that 'r' is the radius of circle in Z plane.

So ' $0 < r < 1$ ' indicates interior part of unit circle. Thus we can conclude that,

L.H.S. of 's' plane is mapped inside the unit circle.

- (ii) If $\sigma = 0$ then $r = e^0 = 1$

$$\therefore \sigma = 0 \Rightarrow r = 1$$

Now $\sigma = 0$ indicates $j\Omega$ axis and $r = 1$ indicates unit circle. Thus, **$j\Omega$ axis in 's' plane is mapped on the unit circle.**

- (iii) If $\sigma > 0$ then, r is equal to 'e' raise to some constant. That means $r > 1$.

$$\therefore \sigma > 0 \Rightarrow r > 1$$

Now $\sigma > 0$ indicates R.H.S. of 's' plane and ' $r > 1$ ' indicates exterior part of unit circle. Thus, **R.H.S. of 's' plane is mapped outside the unit circle.**

Combining all conditions; this mapping is shown in Fig. 2.5.1.

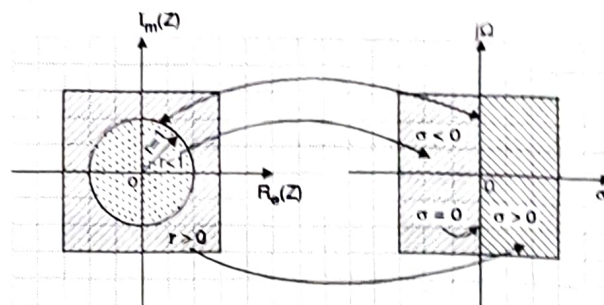


Fig. 2.5.1 : Relationship of s plane to Z-plane

Disadvantages of Impulse Invariance method

1. We know that ' Ω ' is analog frequency and its range is from $\frac{\pi}{T_s}$ to $-\frac{\pi}{T_s}$. While the digital frequency ' ω ' varies from $-\pi$ to π . That means from $\frac{\pi}{T_s}$ to $-\frac{\pi}{T_s}$ ' ω ' maps from $-\pi$ to π . Let k be any integer. Then, we can write the general range of Ω as $(k-1)\frac{\pi}{T_s}$ to $(k+1)\frac{\pi}{T_s}$; but for this range also; ' ω ' maps from $-\pi$ to π . Thus mapping from analog frequency ' Ω ' to digital frequency ' ω ' is many to one. This mapping is not one to one.
2. Analog filters are not band limited so there will be aliasing due to the sampling process. Because of this aliasing, the frequency response of resulting digital filter will not be identical to the original frequency response of analog filter.
3. The change in the value of sampling time (T_s) has no effect on the amount of aliasing.

Some standard formulae for transformation in impulse invariance method are as follows :

$$(i) \frac{1}{s - P_k} \longrightarrow \frac{1}{1 - e^{P_k T_s} z^{-1}}$$

$$r = \frac{1}{e^{\sigma T_s}}$$

$$(ii) \frac{s+a}{(s+a)^2+b^2} \rightarrow \frac{1-e^{-aT_s} [\cos bT_s] Z^{-1}}{1-2e^{-aT_s} [\cos bT_s] Z^{-1} + e^{-2aT_s} \cdot Z^{-2}}$$

$$(iii) \frac{b}{(s+a)^2+b^2} \rightarrow \frac{e^{-aT_s} [\sin bT_s] Z^{-1}}{1-2e^{-aT_s} [\cos bT_s] Z^{-1} + e^{-2aT_s} \cdot Z^{-2}}$$

Design steps for Impulse Invariance method

- ▶ **Step I :** Analog frequency transfer function $H(s)$ will be given. If it is not given then, obtain expression of $H(s)$ from the given specifications.
- ▶ **Step II :** If required expand $H(s)$ by using Partial Fraction Expansion (PFE).
- ▶ **Step III :** Obtain Z transform of each PFE term using impulse invariance transformation equation.
- ▶ **Step IV :** Obtain $H(Z)$, this is required digital IIR filter.

2.5.1 Solved Problems on Impulse Invariance Method

Ex. 2.5.1 MU- May 18, May 19, 7 Marks

Find out $H(Z)$ using impulse invariance method at 5 Hz sampling frequency from $H(s)$ as given below :

$$H(s) = \frac{2}{(s+1)(s+2)}$$

✓ **Soln. :**

- ▶ **Step I :** Given analog transfer function is,

$$H(s) = \frac{2}{(s+1)(s+2)} \quad \dots(1)$$

- ▶ **Step II :** We will expand $H(s)$ using partial fraction expansion as :

$$\therefore H(s) = \frac{A_1}{(s+1)} + \frac{A_2}{(s+2)} \quad \dots(2)$$

Thus poles are at $P_1 = -1$ and $P_2 = -2$.

Now values of A_1 and A_2 are calculated as follows :

$$A_1 = (s - P_1) H(s) \Big|_{s=P_1}$$

$$\therefore A_1 = (s+1) \cdot \frac{2}{(s+1)(s+2)} \Big|_{s=-1}$$

$$\therefore A_1 = \frac{2}{-1+2} = 2$$

$$\text{and } A_2 = (s - P_2) H(s) \Big|_{s=P_2}$$

$$= (s+2) \cdot \frac{2}{(s+1)(s+2)} \Big|_{s=-2}$$

$$\therefore A_2 = \frac{2}{-2+1} = -2$$

Putting values of A_1 and A_2 in Equation (2) we get,

$$H(s) = \frac{2}{(s+1)} - \frac{2}{(s+2)} \quad \dots(3)$$

- ▶ **Step III :** Now we will obtain the Z-transform using impulse invariance transformation equation. It is,

$$\frac{1}{s - P_k} \rightarrow \frac{1}{1 - e^{P_k T_s} \cdot Z^{-1}} \quad \dots(4)$$

Here T_s = Sampling time. Now given sampling frequency is $F_s = 5$ Hz.

$$\therefore T_s = \frac{1}{F_s} = \frac{1}{5} = 0.2 \text{ sec.}$$

We have poles at $P_1 = -1$ and $P_2 = -2$

So using Equation (4) we get,

$$\frac{1}{s+1} \rightarrow \frac{1}{1 - e^{-1(0.2)} \cdot Z^{-1}} = \frac{1}{1 - e^{-0.2} \cdot Z^{-1}} \quad \dots(5)$$

$$\text{and } \frac{1}{s+2} \rightarrow \frac{1}{1 - e^{-2(0.2)} \cdot Z^{-1}} = \frac{1}{1 - e^{-0.4} \cdot Z^{-1}} \quad \dots(6)$$

- ▶ **Step IV :** The transfer function of digital filter is given by,

$$H(Z) = \sum_{k=1}^N \frac{A_k}{1 - e^{P_k T_s} \cdot Z^{-1}}$$

In this case we get,

$$H(Z) = \frac{A_1}{1 - e^{P_1 T_s} \cdot Z^{-1}} + \frac{A_2}{1 - e^{P_2 T_s} \cdot Z^{-1}} \quad \dots(7)$$

Using Equations (5) and (6) we get,

$$\begin{aligned} H(Z) &= \frac{2}{1 - e^{-0.2} \cdot Z^{-1}} - \frac{2}{1 - e^{-0.4} \cdot Z^{-1}} \\ &= \frac{2}{1 - 0.818 Z^{-1}} - \frac{2}{1 - 0.67 Z^{-1}} \end{aligned}$$

To convert each term into positive powers of Z ; multiplying numerator and denominator of each term by Z we get,

$$H(Z) = \frac{2Z}{Z - 0.818} - \frac{2Z}{Z - 0.67}$$

$$\therefore H(Z) = \frac{2Z(Z - 0.67) - 2Z(Z - 0.818)}{(Z - 0.818)(Z - 0.67)}$$

$$\therefore H(Z) = \frac{2Z^2 - 1.34Z - 2Z^2 + 1.636Z}{Z^2 - 0.67Z - 0.818Z + 0.54}$$

$$\therefore H(Z) = \frac{0.29 Z}{Z^2 - 1.488 Z + 0.54}$$

This is the required transfer function for digital IIR filter.

UEx. 2.5.2 MU - May 14, 5 Marks

Obtain a digital filter transfer function by applying impulse invariance transformation on the analog TF.

$$H_a(s) = \frac{s}{s^2 + 3s + 2}$$

Use $F_s = 1 \text{ k samples/sec}$.

✓ **Soln. :**

► **Step I :** Given,

$$H_a(s) = \frac{s}{s^2 + 3s + 2}$$

$$\therefore H_a(s) = \frac{s}{(s+1)(s+2)}$$

► **Step II :** In P.F.E., we can write,

$$H_a(s) = \frac{A_1}{s+1} + \frac{A_2}{s+2}$$

$$\begin{aligned} \text{Here } A_1 &= (s+1) \cdot \frac{s}{(s+1)(s+2)} \Big|_{s=-1} \\ &= \frac{-1}{-1+2} = -1 \end{aligned}$$

$$\begin{aligned} \text{and } A_2 &= (s+2) \cdot \frac{s}{(s+1)(s+2)} \Big|_{s=-2} \\ &= \frac{-2}{-2+1} = 2 \end{aligned}$$

$$\therefore H_a(s) = \frac{-1}{s+1} + \frac{2}{s+2}$$

► **Step III :** The transformation equation is,

$$\frac{1}{s - P_k} \rightarrow \frac{1}{1 - e^{P_k T_s} \cdot Z^{-1}}$$

$$\text{Here } T_s = \frac{1}{F_s} = \frac{1}{1 \text{ k samples/sec}} = \frac{1}{1000} = 10^{-3} \text{ sec}$$

$$\text{and } P_1 = -1, P_2 = -2$$

$$\therefore \frac{1}{s+1} \rightarrow \frac{1}{1 - e^{-1 \times 10^{-3}} \cdot Z^{-1}} = \frac{1}{1 - 0.999 Z^{-1}}$$

$$\text{and } \frac{1}{s+2} \rightarrow \frac{1}{1 - e^{-2 \times 10^{-3}} \cdot Z^{-1}} = \frac{1}{1 - 0.998 Z^{-1}}$$

► **Step IV :** The digital transfer function is given by,

$$H(Z) = \sum_{k=1}^N \frac{A_k}{1 - e^{P_k T_s} \cdot Z^{-1}}$$

$$\therefore H(Z) = \frac{A_1}{1 - 0.999 Z^{-1}} + \frac{A_2}{1 - 0.998 Z^{-1}}$$

$$\therefore H(Z) = \frac{-1}{1 - 0.999 Z^{-1}} + \frac{2}{1 - 0.998 Z^{-1}}$$

► **Ex. 2.5.3 :** Consider a causal continuous time system with impulse response $h_c(t)$ and the system function

$$H_c(s) = \frac{s+a}{(s+a)^2 + b^2}$$

Use impulse invariance to determine $H(Z)$ for a discrete time system such that $h(n) = h_c(nT)$.

✓ **Soln. :**

► **Step I :** Given $H_c(s) = \frac{s+a}{(s+a)^2 + b^2}$

We will obtain roots of $H_c(s)$ as follows ;

$$H_c(s) = \frac{s+a}{(s+a+jb)(s+a-jb)}$$

Here roots, that means poles are at,

$$P_1 = -a-jb, P_2 = -a+jb$$

► **Step II :** In partial fraction expansion form we can write,

$$H_c(s) = \frac{A_1}{(s+a+jb)} + \frac{A_2}{(s+a-jb)} \quad \dots(1)$$

$$\text{Here } A_1 = (s+a+jb) \cdot \frac{s+a}{(s+a+jb)(s+a-jb)} \Big|_{s=-a-jb}$$

$$\therefore A_1 = \frac{-a-jb+a}{-a-jb+a-jb} = \frac{-jb}{-2jb} = \frac{1}{2}$$

$$\text{Similarly, } A_2 = (s+a-jb) \cdot \frac{s+a}{(s+a+jb)(s+a-jb)} \Big|_{s=-a+jb}$$

$$\therefore A_2 = \frac{-a+jb+a}{-a+jb+a-jb} = \frac{jb}{2jb} = \frac{1}{2}$$

Putting A_1 and A_2 in Equation (1) we get,

$$H_c(s) = \frac{1}{2} \cdot \frac{1}{(s+a+jb)} + \frac{1}{2} \cdot \frac{1}{(s+a-jb)} \quad \dots(2)$$

► **Step III :** We have impulse invariance transformation equation,

$$\frac{1}{s - P_k} \rightarrow \frac{1}{1 - e^{P_k T_s} \cdot Z^{-1}}$$

Thus poles P_1 and P_2 are transformed as follows,

$$\frac{1}{s - (-a - jb)} \rightarrow \frac{1}{1 - e^{(-a - jb) T_s} \cdot Z^{-1}}$$

and $\frac{1}{s - (-a + jb)} \rightarrow \frac{1}{1 - e^{(-a + jb) T_s} \cdot Z^{-1}}$

Using this transformation, $H_c(s)$ is converted into $H(Z)$ as follows.

$$\begin{aligned} H(Z) &= \frac{1}{2} \cdot \frac{1}{1 - e^{(-a - jb) T_s} \cdot Z^{-1}} + \frac{1}{2} \cdot \frac{1}{1 - e^{(-a + jb) T_s} \cdot Z^{-1}} \\ \therefore H(Z) &= \frac{1}{2} \left[\frac{1 - e^{(-a + jb) T_s} \cdot Z^{-1} + 1 - e^{(-a - jb) T_s} \cdot Z^{-1}}{(1 - e^{(-a - jb) T_s} \cdot Z^{-1})(1 - e^{(-a + jb) T_s} \cdot Z^{-1})} \right] \\ \therefore H(Z) &= \frac{1}{2} \left[\frac{2 - e^{-a T_s} \cdot e^{jb T_s} \cdot Z^{-1} - e^{-a T_s} \cdot e^{-jb T_s} \cdot Z^{-1}}{(1 - e^{(-a + jb) T_s} \cdot Z^{-1} - e^{(-a - jb) T_s} \cdot Z^{-1} + e^{(-a + jb) T_s} \cdot e^{(-a + jb) T_s} \cdot Z^{-2})} \right] \\ \therefore H(Z) &= \frac{1}{2} \left[\frac{2 - e^{-a T_s} \cdot Z^{-1} (e^{jb T_s} + e^{-jb T_s})}{(1 - e^{-a T_s} \cdot e^{jb T_s} \cdot Z^{-1} - e^{-a T_s} \cdot e^{-jb T_s} \cdot Z^{-1} + e^{-a T_s} \cdot e^{-jb T_s} \cdot e^{-a T_s} \cdot e^{jb T_s} \cdot Z^{-2})} \right] \\ \therefore H(Z) &= \frac{1}{2} \left[\frac{2 - e^{-a T_s} \cdot Z^{-1} (e^{jb T_s} + e^{-jb T_s})}{1 - e^{-a T_s} \cdot Z^{-1} (e^{jb T_s} + e^{-jb T_s}) + e^{-2a T_s} \cdot Z^{-2}} \right] \end{aligned}$$

Now, we have an identity

$$2 \cos \theta = e^{j\theta} + e^{-j\theta}$$

$$\therefore e^{jb T_s} + e^{-jb T_s} = 2 \cos b T_s$$

$$\therefore H(Z) = \frac{1}{2} \left[\frac{2 - e^{-a T_s} \cdot Z^{-1} \cdot 2 \cos b T_s}{1 - e^{-a T_s} \cdot Z^{-1} \cdot 2 \cos b T_s + e^{-2a T_s} \cdot Z^{-2}} \right]$$

$$\therefore H(Z) = \frac{1 - e^{-a T_s} \cdot Z^{-1} \cos b T_s}{1 - 2 e^{-a T_s} \cdot Z^{-1} \cos b T_s + e^{-2a T_s} \cdot Z^{-2}}$$

Ex. 2.5.4 MU- May 2017, 6 Marks

The system function of analog filter is given by,

$$H_a(s) = \frac{s + 0.1}{(s + 0.1)^2 + 9}$$

Design IIR digital filter using impulse invariance method.

✓ **Soln. :**

$$\text{We have designed IIR filter for transfer function of analog filter } H(s) = \frac{s + a}{(s + a)^2 + b^2}$$

Here $a = 0.1$ and $b = 3$.

$$H(Z) = \frac{1 - e^{-0.1 T_s} \cdot Z^{-1} \cos 3 T_s}{1 - 2 e^{-0.1 T_s} \cdot Z^{-1} \cos 3 T_s + e^{-0.2 T_s} \cdot Z^{-2}}$$

Ex. 2.5.5 : Determine $H(Z)$ using impulse invariance method for the system function,

$$H(s) = \frac{1}{(s+0.5)(s^2+0.5s+2)}$$

✓ Soln. :

Step I : The given transfer function is,

$$H(s) = \frac{1}{(s+0.5)(s^2+0.5s+2)} \quad \dots(1)$$

Step II : In the partial fraction expansion form $H(s)$ can be written as,

$$\begin{aligned} H(s) &= \frac{1}{(s+0.5)(s^2+0.5s+2)} \\ &= \frac{A}{(s+0.5)} + \frac{Bs+C}{(s^2+0.5s+2)} \end{aligned} \quad \dots(2)$$

Let us obtain the values of A, B and C.

$$\frac{1}{(s+0.5)(s^2+0.5s+2)} = \frac{A(s^2+0.5s+2) + (Bs+C)(s+0.5)}{(s+0.5)(s^2+0.5s+2)}$$

$$\therefore A(s^2+0.5s+2) + (Bs+C)(s+0.5) = 1$$

$$\therefore As^2 + A(0.5s) + 2A + Bs^2 + B(0.5s) + Cs + 0.5C = 1$$

$$s^2(A+B) + s(0.5A+0.5B+C) + (2A+0.5C) = 1 \quad \dots(3)$$

Now s^2 term is absent in R.H.S.

$$\therefore A+B = 0 \quad \dots(4)$$

Similarly 's' term is absent in R.H.S.

$$\therefore 0.5A + 0.5B + C = 0 \quad \dots(5)$$

$$\text{And } 2A + 0.5C = 1 \quad \dots(6)$$

Now we will solve Equations (4), (5) and (6) to obtain the values of A, B and C.

From Equation (4),

$$B = -A$$

Putting this value in Equation (5) we get,

$$0.5A - 0.5A + C = 0$$

$$\therefore C = 0$$

From Equation (6),

$$2A + 0 = 1$$

$$\therefore A = 0.5$$

Since $B = -A$ we get,

$$\therefore B = -0.5$$

Putting these values in Equation (2) we get,

$$H(s) = \frac{0.5}{s+0.5} - \frac{0.5s}{s^2+0.5s+2} \quad \dots(7)$$

To use the standard transformation formulae we will convert the second term on R.H.S. in the form $\frac{s+a}{(s+a)^2+b^2}$ and $\frac{b}{(s+a)^2+b^2}$

Consider the term $s^2 + 0.5s + 2$. It can be expressed as,

$$s^2 + 0.5s + 2 = (s^2 + 0.5s + 0.0625) + (1.9375)$$

$$\therefore s^2 + 0.5s + 2 = (s + 0.25)^2 + (1.39)^2$$

Putting this value in Equation (7) we get,

$$H(s) = \frac{0.5}{s + 0.5} - \frac{0.5s}{(s + 0.25)^2 + (1.39)^2}$$

$$\therefore H(s) = \frac{0.5}{s + 0.5} - 0.5 \left[\frac{s + 0.25 - 0.25}{(s + 0.25)^2 + (1.39)^2} \right]$$

$$\therefore H(s) = \frac{0.5}{s + 0.5} - 0.5 \left[\frac{s + 0.25}{(s + 0.25)^2} \right] - 0.5 \left[\frac{-0.25}{(s + 0.25)^2 + (1.39)^2} \right]$$

Now we want the numerator of third term equal to 1.39. It is arranged as follows,

$$H(s) = \frac{0.5}{s + 0.5} - 0.5 \left[\frac{s + 0.25}{(s + 0.25)^2} \right] + \frac{0.5 \times 0.25}{1.39} \left[\frac{1.39}{(s + 0.25)^2 + (1.39)^2} \right]$$

$$\therefore H(s) = \frac{0.5}{s + 0.5} - 0.5 \left[\frac{s + 0.25}{(s + 0.25)^2} \right] + 0.089 \left[\frac{1.39}{(s + 0.25)^2 + (1.39)^2} \right] \quad \dots(8)$$

Now recall the standard transformation formulae,

$$(i) \frac{1}{s - P_k} \rightarrow \frac{1}{1 - e^{P_k T_s} \cdot Z^{-1}}$$

$$(ii) \frac{s + a}{(s + a)^2 + b^2} \rightarrow \frac{1 - e^{-aT_s} [\cos bT_s] Z^{-1}}{1 - 2e^{-aT_s} [\cos bT_s] Z^{-1} + e^{-2aT_s} \cdot Z^{-2}}$$

$$(iii) \frac{b}{(s + a)^2 + b^2} \rightarrow \frac{e^{-aT_s} [\sin bT_s] Z^{-1}}{1 - 2e^{-aT_s} [\cos bT_s] Z^{-1} + e^{-2aT_s} \cdot Z^{-2}}$$

► **Step III :** Using these formulae, equation of $H(Z)$ can be obtained from Equation (8) as follows :

$$H(Z) = \frac{0.5}{1 - e^{0.5T_s} \cdot Z^{-1}} - 0.5 \left[\frac{1 - e^{-0.25T_s} [\cos 1.39T_s] Z^{-1}}{1 - 2e^{-0.25T_s} [\cos 1.39T_s] Z^{-1} + e^{-0.5T_s} \cdot Z^{-2}} \right] + 0.089 \left[\frac{1 - e^{-0.25T_s} [\sin 1.39T_s] Z^{-1}}{1 - 2e^{-0.25T_s} [\cos 1.39T_s] Z^{-1} + e^{-0.5T_s} \cdot Z^{-2}} \right]$$

This is the required transfer function. Note that the value of T_s is not given in the problem; so T_s is kept as it is.

Ex. 2.5.6 : Convert an analog filter with system function $H(s)$ into digital IIR filter using impulse-invariance method.

$$H(s) = \frac{10}{s^2 + 7s + 10}$$

✓ **Soln. :**

$$H(s) = \frac{10}{s^2 + 7s + 10}$$

$$\therefore H(s) = \frac{10}{(s + 5)(s + 2)}$$

$$\therefore H(s) = \frac{A_1}{s + 5} + \frac{A_2}{s + 2}$$

Assume
 $T = 1 \text{ sec}$

$$\text{Here } A_1 = (s + 5) \cdot \frac{10}{(s + 5)(s + 2)} \Big|_{s = -5} = \frac{10}{-3}$$

$$\text{and } A_2 = (s + 2) \cdot \frac{10}{(s + 5)(s + 2)} \Big|_{s = -2} = \frac{10}{3}$$

$$\therefore H(s) = \frac{-10/3}{s + 5} - \frac{10/3}{s + 2}$$

$$\therefore H(s) = \frac{-10/3}{s - (-5)} - \frac{10/3}{s - (-2)}$$

Thus poles are at $P_1 = 5$ and $P_2 = 2$.

We have Impulse invariance transformation equation,

$$\frac{1}{s - p_k} \rightarrow \frac{1}{1 - e^{p_k T_s} \cdot Z^{-1}}$$

$$\text{Let } T_s = 1$$

$$\frac{1}{s - (-5)} \rightarrow \frac{1}{1 - e^{-5} \cdot Z^{-1}} = \frac{1}{1 - 6.74 \times 10^{-3} \cdot Z^{-1}}$$

$$\text{and } \frac{1}{s - (-2)} \rightarrow \frac{1}{1 - e^{-2} \cdot Z^{-1}} = \frac{1}{1 - 0.135 \cdot Z^{-1}}$$

$$\therefore H(Z) = \frac{-10/3}{1 - 6.74 \times 10^{-3} \cdot Z^{-1}} + \frac{10/3}{1 - 0.135 Z^{-1}}$$

UEx. 2.5.7 MU - Q. 2(b), Dec. 2019, 5 Marks

Transform analog filter transfer function $H(s)$ given below in to digital filter transfer function $H(z)$ using Impulse Invariance Transformation method with $T = 1$. $H(s) = \frac{0.5(s+4)}{(s+1)(s+2)}$

✓ **Soln. :**

$$\begin{aligned} \text{Given, } H(s) &= \frac{0.5(s+4)}{(s+1)(s+2)} \\ &= \frac{A_1}{s+1} + \frac{A_2}{s+2} \end{aligned}$$

$$\text{Here, } A_1 = \left. \frac{0.5(s+4)}{(s+2)} \right|_{s=-1} = \frac{0.5(-1+4)}{-1+2} = 1.5$$

$$A_2 = \left. \frac{0.5(s+4)}{(s+1)} \right|_{s=-2} = \frac{0.5(-2+4)}{-2+1} = -1$$

Using impulse invariance transformation equation

$$\frac{1}{s - p_k} = \frac{1}{1 - e^{p_k T_s} z^{-1}}$$

$$\therefore \frac{1}{s+1} = \frac{1}{1 - e^{-1.1} z^{-1}} = \frac{1}{1 - 0.3678 z^{-1}}$$

$$\text{and } \frac{1}{s+2} = \frac{1}{1 - e^{-2.1} z^{-1}} = \frac{1}{1 - 0.1353 z^{-1}}$$

$$\therefore H(z) = \frac{1.5}{1 - 0.3678 z^{-1}} - \frac{1}{1 - 0.1353 z^{-1}}$$

Syllabus Topic : Bilinear Transformation Method**2.5.2 Bilinear Transformation Method (BLT)**

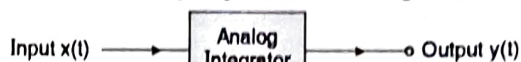
UQ. Using trapezoidal rule of integration show that

$$S = \frac{2(1-Z^{-1})}{T(1+Z^{-1})} \quad (\text{MU - Dec. 04, May 06, 5 Marks})$$

UQ. Obtain the mapping formula for the bilinear transformation method. **MU - Dec. 09, 5 Marks**

UQ. Explain the frequency warping in Bilinear transformation. **MU - May 14, 5 Marks**

- In case of impulse invariance method, we have studied that the mapping is many to one. So this method is not suitable to design high-pass filter and band reject filter.
- In case of bilinear transformation; the mapping is one to one from s domain to the Z domain. So there is no aliasing effect. The limitations of impulse invariance method are overcome by using BLT method.
- Consider an analog integrator as shown in Fig. 2.5.2(a).



$$H_n(s) = \frac{1}{s}$$

Fig. 2.5.2(a)

- The response of analog integrator is,

$$H_n(s) = \frac{1}{s} \quad \dots(2.5.16)$$

- Input in laplace domain is $X(s)$ and output in laplace domain is $Y(s)$.
- For time period T ; the difference in output is given by,

$$y(t_2) - y(t_1) = \int_{t_1}^{t_2} x(n T_s) dT \quad \dots(2.5.17)$$

- Consider input signal as shown in Fig. 2.5.2(b).
- Here we have assumed two input positions $x(t_1)$ and $x(t_2)$. Corresponding output is denoted by $y(t_1)$ and $y(t_2)$ respectively.
- Now area under the curve is addition of area of triangle ABC and area of rectangle ACDE.

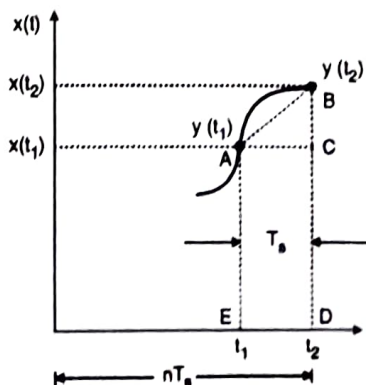


Fig. 2.5.2(b)

$$\therefore y(t_2) - y(t_1) = \frac{1}{2} (t_2 - t_1) [x(t_2) + x(t_1)] \quad \dots(2.5.18)$$

- As shown in Fig. 2.5.2(b), time period $t_2 = n T_s$. Thus time period t_1 is $n T_s - T_s$ that means $t_2 - t_1 = T_s$

- Putting these values in Equation (2.5.18) we get,

$$y(n T_s) - y(n T_s - T_s) = \frac{1}{2} T [x(n T_s) - x(n T_s - T)] + x(n T_s - T) T_s$$

$$y(n T_s) - y(n T_s - T_s) = \frac{1}{2} T_s x(n T) - \frac{1}{2} T_s x(n T_s - T_s) + T_s x(n T_s - T_s)$$

$$y(n T_s) - y(n T_s - T_s) = \frac{1}{2} T_s x(n T_s) + \frac{1}{2} T_s x(n T_s - T_s)$$

$$y(n T_s) - y(n T_s - T_s) = \frac{T_s}{2} [x(n T_s) + x(n T_s - T_s)] \quad \dots(2.5.19)$$

- Taking Z transform of both sides we get,

$$Y(Z) - Z^{-1} Y(Z) = \frac{T_s}{2} [X(Z) + Z^{-1} X(Z)]$$

$$\therefore Y(Z) [1 - Z^{-1}] = \frac{T_s}{2} [X(Z) (1 + Z^{-1})]$$

$$\therefore \frac{Y(Z)}{X(Z)} = H(Z) = \frac{T_s}{2} \frac{(1 + Z^{-1})}{(1 - Z^{-1})} \quad \dots(2.5.20)$$

- Now we have transfer function of analog filter,

$$H_a(s) = \frac{1}{s} \quad \dots(2.5.21)$$

- Equating Equations (2.5.20) and (2.5.21) we get,

$$\frac{1}{s} = \frac{T_s}{2} \frac{(1 + Z^{-1})}{(1 - Z^{-1})} \quad \dots(2.5.22)$$

- Thus relationship between s plane and Z-plane is given by,

$$s = \frac{2}{T_s} \left(\frac{Z-1}{Z+1} \right) \quad \dots(2.5.23)$$

Here T_s is the sampling time.

- We know that 's' is the Laplace operator and it can be expressed as,

$$s = \sigma + j\Omega \quad \dots(2.5.24)$$

- Now the equation of Z in polar form is,

$$Z = r e^{j\omega} \quad \dots(2.5.25)$$

- Putting Equations (2.5.24) and (2.5.25) in Equation (2.5.23) we get,

- Multiplying numerator and denominator by $(r e^{-j\omega} + 1)$ we get,

$$\sigma + j\Omega = \frac{2}{T_s} \left[\frac{r e^{j\omega} - 1}{r e^{j\omega} + 1} \right] \times \frac{r e^{-j\omega} + 1}{r e^{-j\omega} + 1}$$

$$\sigma + j\Omega = \frac{2}{T_s} \left[\frac{r^2 e^{j\omega} \cdot e^{-j\omega} + r e^{j\omega} - r e^{-j\omega} - 1}{r^2 e^{j\omega} \cdot e^{-j\omega} + r e^{j\omega} + r e^{-j\omega} + 1} \right] \quad \dots(2.5.26)$$

$$\text{But } e^{j\omega} \cdot e^{-j\omega} = e^0 = 1$$

$$\therefore \sigma + j\Omega = \frac{2}{T_s} \left[\frac{r^2 + r e^{j\omega} - r e^{-j\omega} - 1}{r^2 + r e^{j\omega} + r e^{-j\omega} + 1} \right] \quad \dots(2.5.27)$$

$$\text{Now we have, } \frac{e^{j\omega} - e^{-j\omega}}{2j} = \sin \omega \quad \text{and} \quad \frac{e^{j\omega} + e^{-j\omega}}{2} = \cos \omega$$

- We will rearrange Equation (2.5.27) as follows :

$$\sigma + j\Omega = \frac{2}{T_s} \left[\frac{r^2 - 1 + j2r \left(\frac{e^{j\omega} - e^{-j\omega}}{2j} \right)}{r^2 + 2r \left(\frac{e^{j\omega} + e^{-j\omega}}{2} \right) + 1} \right]$$

$$\therefore \sigma + j\Omega = \frac{2}{T_s} \left[\frac{r^2 - 1 + j2r \sin \omega}{r^2 + 2r \cos \omega + 1} \right]$$

$$\therefore \sigma + j\Omega = \frac{2}{T_s} \left[\frac{r^2 - 1}{r^2 + 2r \cos \omega + 1} + j \frac{2r \sin \omega}{r^2 + 2r \cos \omega + 1} \right] \quad \dots(2.5.28)$$

- Equating real and imaginary parts of Equation (2.5.28) we get,

$$\sigma = \frac{2}{T_s} \times \frac{r^2 - 1}{r^2 + 2r \cos \omega + 1} \quad \dots(2.5.29)$$

$$\text{and } \Omega = \frac{2}{T_s} \times \frac{2r \sin \omega}{r^2 + 2r \cos \omega + 1} \quad \dots(2.5.30)$$

- Now we will discuss the following conditions related to Equation (2.5.29).

(i) When $r < 1$ then $\sigma < 0$

Here $r < 1$, means interior part of circle having unit circle and $\sigma < 0$, means σ is negative which is L.H.S. of s-plane. So this condition indicates that L.H.S. of s plane maps inside the unit circle.

(ii) When $r = 1$ then $\sigma = 0$

Now $r = 1$ means unit circle and $\sigma = 0$ means $j\Omega$ axis. Thus this condition indicates that the $j\Omega$ axis maps on the unit circle.

(iii) When $r > 1$ then $\sigma > 0$

Here $r > 1$, means exterior part of unit circle and $\sigma > 0$ indicates that σ is positive means R.H.S. of s-plane. So this condition indicates that R.H.S. of s-plane maps outside the unit circle.

This mapping is similar to the mapping in impulse invariance method, as shown in Fig. 2.5.3. But in impulse invariance method mapping is valid only for poles ; while in bilinear transformation, mapping is valid for poles as well as zeros.

How stable analog filter is converted into stable digital filter ?

Analog filter is stable if the poles lie on the L.H.S. of s-plane. While the digital filter is stable if the poles are inside the unit circle in the Z-domain. Now condition (i) indicates that L.H.S. of s-plane maps inside the unit circle. Thus stable analog filter is converted into stable digital filter.

2.5.3 Frequency Warping Concept

UQ. Using trapezoidal of integration show that $\Omega = \frac{2}{T_s} \tan \frac{\omega}{2}$. Explain frequency warping effect.

MU - Dec. 04, May 06, Dec. 06, May 15, 5 Marks

- Here we will obtain the relationship of $j\Omega$ axis in s-plane to the unit circle in the Z-plane ($r = 1$).

Recall Equation (2.5.23).

$$\Omega = \frac{2}{T_s} \times \frac{2r \sin \omega}{r^2 + 2r \cos \omega + 1}$$

- For the unit circle, $r = 1$. Thus putting $r = 1$ in the equation of Ω we get,

$$\Omega = \frac{2}{T_s} \times \frac{2 \sin \omega}{1 + 2 \cos \omega + 1}$$

$$\therefore \Omega = \frac{2}{T_s} \times \frac{2 \sin \omega}{2 + 2 \cos \omega}$$

$$\therefore \Omega = \frac{2}{T_s} \times \frac{\sin \omega}{1 + \cos \omega} \quad \dots(2.5.31)$$

We have the trigonometric identities,

$$\sin \omega = 2 \sin \frac{\omega}{2} \cdot \cos \frac{\omega}{2} \text{ and } 2 \cos^2 \frac{\omega}{2} = 1 + \cos \omega.$$

Thus Equation (2.5.31) becomes,

$$\Omega = \frac{2}{T_s} \times \frac{2 \sin \frac{\omega}{2} \cos \frac{\omega}{2}}{2 \cos^2 \frac{\omega}{2}}$$

$$\therefore \Omega = \frac{2}{T_s} \times \frac{2 \sin \frac{\omega}{2}}{2 \cos \frac{\omega}{2}}$$

$$\therefore \Omega = \frac{2}{T_s} \tan \frac{\omega}{2}$$

$$\therefore \omega = 2 \tan^{-1} \left(\frac{\Omega T_s}{2} \right) \quad \dots(2.5.32)$$

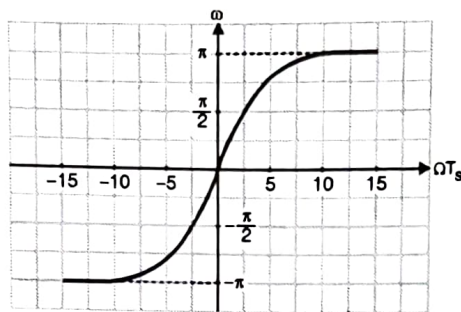


Fig. 2.5.3 : Mapping between ω and Ω

Now for different values of ΩT_s ; the graph of ΩT_s versus ω is as shown in Fig. 2.5.3.

Observations

- The entire range in Ω is mapped only once into the range $-\pi \leq \omega \leq \pi$.
- The mapping is one to one.
- The mapping is highly non-linear.

Now because of the non-linearity of tangent function $2 \tan^{-1} \left(\frac{\Omega T_s}{2} \right)$; there exists **frequency warping or frequency compression**.

What is frequency warping ?

Because of the non-linear mapping; the amplitude response of digital IIR filter is expanded at lower frequencies and compressed at higher frequencies in comparison to the analog filter. This effect is called as frequency warping.

2.5.4 Prewarping Procedure

We have discussed the warping effect. Because of this effect, there is non-linear compression of Ω to ω values. To compensate this effect; prewarping or prescaling procedure is used. This procedure is as follows :

- The ' Ω ' scale is changed to prewarped scale denoted by Ω^* and $\Omega^* = \frac{2}{T_s} \tan \left(\frac{\omega}{2} \right)$ $\Omega^* = \frac{2}{T_s} \tan \left(\frac{\omega T_s}{2} \right)$
- Then analog filter transfer function $H(s)$ is obtained using values of Ω^* .
- By applying the bilinear transformation, the desired digital frequency response $H(Z)$ is obtained.

2.5.5 Advantages of Bilinear Transformation Method

1. There is one to one transformation from the s-domain to the Z-domain.
2. The mapping is one to one.
3. There is no aliasing effect.
4. Stable analog filter is transformed into the stable digital filter.

2.5.6 Disadvantage of Bilinear Transformation Method

The mapping is non-linear and because of this; frequency warping effect takes place.

2.5.7 Comparison between Impulse Invariance and Bilinear Transformation Method

UQ. Compare bilinear transformation and impulse invariance methods.

MU - May 11, 4 Marks, May 18, 5 Marks

UQ. Compare impulse invariance method and BLT method.

MU - May 19, 5 Marks

Sr. No.	Impulse invariance method	Bilinear transformation method
1.	Poles are transferred by using the equation, $\frac{1}{s - P_k} \rightarrow \frac{1}{1 - e^{P_k T_s} \cdot Z^{-1}}$	Poles are transferred by using the equation. $s = \frac{2}{T_s} \frac{Z-1}{Z+1}$
2.	Mapping is many to one.	Mapping is one to one.
3.	Aliasing effect is present.	Aliasing effect is not present.
4.	It is not suitable to design high-pass filter and band reject filter.	High pass filter and band reject filter can be designed.
5.	Only poles of the system can be mapped.	Poles as well as zeros can be mapped.
6.	No frequency warping effect.	Frequency warping effect is present

2.5.8 Solved Problems on BLT

Ex. 2.5.8 : An analog filter has the following transfer function $H(s) = \frac{1}{s+1}$. Using bilinear transformation technique, determine the transfer function of digital filter $H(Z)$ and also write the difference equation of digital filter.

✓ Soln. :

The given transfer function is,

$$H(s) = \frac{1}{s+1}$$

In bilinear transformation $H(Z)$ is obtained by putting,

$$s = \frac{2}{T_s} \left[\frac{Z-1}{Z+1} \right]$$

Here T_s is the sampling time; which is not given. So $T_s = 1$ sec.

$$\therefore s = 2 \left[\frac{Z-1}{Z+1} \right]$$

Putting this value in Equation (1),

$$H(Z) = \frac{1}{1 + 2 \left(\frac{Z-1}{Z+1} \right)} = \frac{1}{\frac{(Z+1) + 2(Z-1)}{Z+1}}$$

$$\therefore H(Z) = \frac{Z+1}{Z+1+2Z-2} = \frac{Z+1}{3Z-1}$$

This is the transfer function of digital filter. Now we have,

$$H(Z) = \frac{Y(Z)}{X(Z)} = \frac{Z+1}{3Z-1}$$

Converting into negative power of Z we get,

$$\frac{Y(Z)}{X(Z)} = \frac{1+Z^{-1}}{3-Z^{-1}}$$

$$\therefore Y(Z)(3-Z^{-1}) = X(Z)(1+Z^{-1})$$

$$\therefore 3Y(Z) - Z^{-1} \cdot Y(Z) = X(Z) + Z^{-1}X(Z)$$

Taking IZT of both sides,

$$3y(n) - y(n-1) = x(n) + x(n-1)$$

$$\therefore 3y(n) = x(n) + x(n-1) + y(n-1)$$

$$\therefore y(n) = \frac{1}{3}x(n) + \frac{1}{3}x(n-1) + \frac{1}{3}y(n-1)$$

This is the difference equation of digital filter.

UEx. 2.5.9 MU - May 2016, Dec. 2016, 5 Marks

The transfer function of analog filter is :

$$H(s) = \frac{3}{(s+2)(s+3)} \text{ with } T_s = 0.1 \text{ sec.}$$

Design the digital IIR filter using BLT.

✓ Soln. : The given transfer function is,

$$H(s) = \frac{3}{(s+2)(s+3)}$$

In bilinear transformation, $H(Z)$ is obtained by putting :

$$s = \frac{2}{T_s} \left[\frac{Z-1}{Z+1} \right]$$

Here T_s = sampling time = 0.1 sec.

$$s = \frac{2}{0.1} \left[\frac{Z-1}{Z+1} \right] = 20 \left[\frac{Z-1}{Z+1} \right]$$

Putting this value in Equation (1) we get,

$$H(Z) = \frac{3}{20 \left(\frac{Z-1}{Z+1} \right) + 2} \left[20 \left(\frac{Z-1}{Z+1} \right) + 3 \right]$$

$$\therefore H(Z) = \frac{3}{\left[\frac{20Z-20}{Z+1} + 2 \right] \left[\frac{20Z-20}{Z+1} + 3 \right]}$$

$$\therefore H(Z) = \frac{3(Z+1)(Z+1)}{(20Z-20+2Z+2)(20Z-20+3Z+3)}$$

$$\therefore H(Z) = \frac{3(Z+1)^2}{(22Z-18)(23Z-17)}$$

$$\therefore H(Z) = \frac{3(Z^2+2Z+1)}{506Z^2-374Z-414Z+306}$$

$$\therefore H(Z) = \frac{3(Z^2+2Z+1)}{506Z^2-788Z+306}$$

$$\therefore H(Z) = \frac{Z^2+2Z+1}{168.67Z^2-262.67Z+102}$$

This is the required transfer function for digital IIR filter.

Ex. 2.5.10 : Design a single pole low pass digital filter with a 3 dB bandwidth of 0.2π by use of bilinear transformation applied to the analog filter.

$H(s) = \frac{\Omega_c}{s + \Omega_c}$, where Ω_c is the 3 dB bandwidth of analog filter.

☒ **Soln. :** The given transfer function of analog filter is,

$$H(s) = \frac{\Omega_c}{s + \Omega_c} \quad \dots(1)$$

The cut-off frequency of digital filter (ω_c) is given. It is,

$$\omega_c = 0.2\pi \quad \dots(2)$$

We know that in BLT, Ω_c and ω_c are related as,

$$\Omega_c = \frac{2}{T_s} \tan \left[\frac{\omega_c}{2} \right]$$

$$= \frac{2}{T_s} \tan \left[\frac{0.2\pi}{2} \right] = \frac{2}{T_s} \tan [0.1\pi]$$

$$\therefore \Omega_c = \frac{0.65}{T_s} \quad \dots(3)$$

Putting this value in Equation (1) we get,

$$H(s) = \frac{\frac{0.65}{T_s}}{s + \frac{0.65}{T_s}} \quad \dots(4)$$

Thus we have obtained the transfer function of analog filter in the appropriate form.

Now to obtain $H(Z)$ we have to put,

$$s = \frac{2}{T_s} \left[\frac{Z-1}{Z+1} \right] \text{ in Equation (4)}$$

$$\therefore H(Z) = \frac{0.65/T_s}{\frac{2}{T_s} \left[\frac{Z-1}{Z+1} \right] + \frac{0.65}{T_s}}$$

Multiplying numerator and denominator by T_s we get,

$$H(Z) = \frac{0.65}{2 \left[\frac{Z-1}{Z+1} \right] + 0.65} = \frac{0.65}{\frac{2Z-2}{Z+1} + 0.65}$$

$$\therefore H(Z) = \frac{0.65(Z+1)}{2Z-2+0.65Z+0.65} = \frac{0.65(Z+1)}{2.65Z-1.35}$$

Dividing numerator and denominator by 2.65 we get,

$$H(Z) = \frac{0.245(Z+1)}{Z-0.509} \quad \dots(5)$$

This is the required transfer function for digital filter. Note that there is a single pole at $P_1 = 0.509$.

Now in the problem frequency of digital filter is given which is $\omega_c = 0.2\pi$ and bandwidth is 3 dB. We will check the frequency response. The frequency response of digital filter is obtained by putting $Z = e^{j\omega}$ in the equation of $H(Z)$. Thus we get,

$$H(\omega) = H(Z) \Big|_{Z=e^{j\omega}}$$

$$\therefore H(\omega) = \frac{0.245(e^{j\omega}+1)}{e^{j\omega}-0.509} \quad \dots(6)$$

If $\omega = 0$ then Equation (6) becomes,

$$H(0) = \frac{0.245(e^0+1)}{e^0-0.509} = \frac{0.49}{0.49} = 1$$

Thus at $\omega = 0$; the magnitude $|H(\omega)| = 1$.

Now at $\omega = 0.2\pi$ we get,

$$H(\omega) = \frac{0.245(e^{j0.2\pi}+1)}{e^{j0.2\pi}-0.509}$$

$$= \frac{0.245[\cos(0.2\pi) + j\sin(0.2\pi) + 1]}{\cos(0.2\pi) + j\sin(0.2\pi) - 0.509}$$

$$= \frac{0.245[0.809 + j0.587 + 1]}{0.809 + j0.587 - 0.509}$$

$$\therefore H(\omega) = \frac{0.245(1.809 + j0.587)}{0.3 + j0.587}$$

$$\therefore |H(\omega)| = 0.707 = 3 \text{ dB}$$

Thus the desired response is obtained.

UEx. 2.5.11 MU - May 05, 4 Marks, May 14, May 15, 8 Marks, Dec. 15, 5 Marks

The system transfer function of analog filter is given by,

$$H(s) = \frac{s+0.1}{(s+0.1)^2 + 16} \quad \text{Obtain the system transfer function of}$$

digital filter using BLT which is resonant at $\omega_r = \frac{\pi}{2}$.

✓ **Soln. :**

The given transfer function is,

$$H(s) = \frac{s+0.1}{(s+0.1)^2 + 16}$$

$$\therefore H(s) = \frac{s+0.1}{(s+0.1)^2 + (4)^2} \quad \dots(1)$$

From Equation (1) we can say that $\Omega = 4$.

The value of ω_r is given as $\omega_r = \frac{\pi}{2}$

Now we will find out the value of sampling time (T_s) using the relation.

$$\Omega = \frac{2}{T_s} \tan\left(\frac{\omega}{2}\right)$$

$$\therefore 4 = \frac{2}{T_s} \tan\left(\frac{\pi}{4}\right)$$

$$\therefore T_s = \frac{2}{4} \tan\left(\frac{\pi}{4}\right)$$

$$\therefore T_s = 0.5 \text{ sec.}$$

Using bilinear transformation $H(Z)$ can be obtained by putting, $s = \frac{2}{T_s} \left(\frac{Z-1}{Z+1} \right)$ in the equation of $H(s)$.

$$\therefore H(Z) = \frac{\frac{2}{0.5} \left(\frac{Z-1}{Z+1} \right) + 0.1}{\left[\frac{2}{0.5} \left(\frac{Z-1}{Z+1} \right) + 0.1 \right]^2 + 16}$$

$$\therefore H(Z) = \frac{4 \left(\frac{Z-1}{Z+1} \right) + 0.1}{\left[4 \left(\frac{Z-1}{Z+1} \right) + 0.1 \right]^2 + 16}$$

$$= \frac{\frac{4Z-4}{Z+1} + 0.1}{\left[\frac{4Z-4}{Z+1} + 0.1 \right]^2 + 16}$$

$$\therefore H(Z) = \frac{\frac{4Z-4}{Z+1} + 0.1}{\left[\frac{4Z-4+0.1Z+0.1}{Z+1} \right]^2 + 16}$$

$$= \frac{\frac{4.1Z-3.9}{Z+1}}{\left[\frac{4.1Z-3.9}{Z+1} \right]^2 + 16}$$

$$\therefore H(Z) = \frac{\frac{4.1Z-3.9}{Z+1}}{\frac{(16.81Z^2 - 31.98Z + 15.21)}{(Z+1)^2} + 16}$$

$$\therefore H(Z) = \frac{\frac{(4.1Z-3.9)}{Z+1} \times (Z+1)^2}{16.81Z^2 - 31.98Z + 15.21 + 16(Z+1)^2}$$

$$\therefore H(Z) = \frac{(4.1Z-3.9)(Z+1)}{16.81Z^2 - 31.98Z + 15.21 + 16Z^2 + 32Z + 16}$$

$$\therefore H(Z) = \frac{4.1Z^2 + 4.1Z - 3.9Z - 3.9}{32.81Z^2 + 0.02Z + 31.21}$$

$$= \frac{4.1Z^2 + 0.2Z - 3.9}{32.81Z^2 + 0.02Z + 31.21}$$

This is the required transfer function for digital IIR filter.

Ex. 2.5.12 : $H_a(s)$ is given as, $H_a(s) = \frac{1}{(s+1)^2}$ and $T_s = 0.1 \text{ sec}$

Find $H(Z)$ using bilinear transform.

✓ **Soln. :** Given,

$$H_a(s) = \frac{1}{(s+1)^2} = \frac{1}{s^2 + 2s + 1}$$

Now $H(Z)$ is obtained by replacing s by $\frac{2}{T_s} \left[\frac{Z-1}{Z+1} \right]$.

Here $T_s = 0.1 \text{ sec}$.

$$\therefore H(Z) = \frac{1}{\left\{ \frac{2}{0.1} \left[\frac{Z-1}{Z+1} \right] \right\}^2 + 2 \left\{ \frac{2}{0.1} \left[\frac{Z-1}{Z+1} \right] \right\} + 1}$$

$$\therefore H(Z) = \frac{1}{400 \left[\frac{Z-1}{Z+1} \right]^2 + 40 \left[\frac{Z-1}{Z+1} \right] + 1}$$

$$\therefore H(Z) = \frac{1}{400 \left[\frac{1-Z^{-1}}{1+Z^{-1}} \right]^2 + 40 \left[\frac{1-Z^{-1}}{1+Z^{-1}} \right] + 1}$$

Ex. 2.5.13 : The analog transfer function of low pass filter is $H(s) = \frac{1}{s+2}$ and its bandwidth is 1 rad/sec. Design the digital filter using BLT method whose cut-off frequency is 20π and sampling time is 0.0167 sec; by considering the warping effect.

✓ **Soln. :** Given analog transfer function is,

$$H(s) = \frac{1}{s+2} \quad \dots(1)$$

For the prewarping procedure we have,

$$\Omega_p^* = \frac{2}{T_s} \tan\left(\frac{\omega T_s}{2}\right)$$

$$\therefore \Omega_p^* = \frac{2}{0.0167} \tan\left(\frac{20\pi \times 0.0167}{2}\right) \dots \text{Here } \omega = 20\pi \text{ (given)}$$

$$\therefore \Omega_p^* = 69.31$$

Now the value of $H^*(s)$ is obtained by putting $s = \frac{s}{\Omega_p^*}$ in

Equation (1).

$$\therefore H^*(s) = \frac{1}{\frac{s}{\Omega_p^*} + 2} = \frac{1}{\frac{s}{69.31} + 2} = \frac{69.31}{s + 138.62}$$

Now $H(Z)$ is obtained by putting $s = \frac{2}{T_s} \left(\frac{Z-1}{Z+1}\right)$ in

Equation (2),

$$\therefore H(Z) = \frac{69.31}{\frac{2}{0.0167} \left(\frac{Z-1}{Z+1}\right) + 138.62}$$

$$= \frac{69.31}{119.76 \left(\frac{Z-1}{Z+1}\right) + 138.62}$$

$$\therefore H(Z) = \frac{69.31(Z+1)}{119.76Z - 119.76 + 138.62Z + 138.62}$$

$$\therefore H(Z) = \frac{69.31(Z+1)}{258.38Z + 18.86} = \frac{Z+1}{3.73Z + 0.27}$$

This is the required transfer function for digital filter.

Ex. 2.5.14 : Design a digital low pass IIR filter to approximate the following transfer function.

$$H(s) = \frac{1}{s^2 + \sqrt{2}s + 1}$$

Using the bilinear transformation method, obtain the transfer function $H(Z)$ of digital filter assuming 3 dB cut-off frequency of 150 Hz and sampling frequency of 1.28 KHz.

☒ **Soln. :** Given,

$$\text{Cut-off frequency } F_p = 150 \text{ Hz,}$$

$$\text{Sampling frequency } F_s = 1.28 \text{ KHz}$$

$$\therefore \omega_p = 2\pi F_p = 300\pi$$

We will consider prewarping effect.

► **Step I :** We have, $\Omega_p^* = \frac{2}{T_s} \tan\left(\frac{\omega_p T_s}{2}\right)$

$$\text{Here } T_s = \frac{1}{F_s} = \frac{1}{1.28 \times 10^3}$$

$$\therefore \Omega_p^* = \frac{2}{1/(1.28 \times 10^3)} \tan\left(\frac{300\pi}{2} \times \frac{1}{1.28 \times 10^3}\right)$$

$$\therefore \Omega_p^* = 987.5$$

► **Step II :** Equation of $H^*(s)$ is obtained by replacing s by $\frac{s}{\Omega_p^*}$ in the equation of $H(s)$.

$$\therefore H^*(s) = \frac{1}{\left(\frac{s}{\Omega_p^*}\right)^2 + \sqrt{2}\left(\frac{s}{\Omega_p^*}\right) + 1}$$

$$\therefore H^*(s) = \frac{1}{\left(\frac{s}{987.5}\right)^2 + \sqrt{2}\left(\frac{s}{987.5}\right) + 1}$$

$$\therefore H^*(s) = \frac{1}{10^{-6} \times 1.025 s^2 + 1.432 \times 10^{-3} s + 1}$$

► **Step III :** $H(Z)$ is obtained by replacing s by $\frac{2}{T_s} \left[\frac{Z-1}{Z+1}\right]$

$$\therefore H(Z) =$$

$$\frac{1}{10^{-6} \times 1.025 \left\{ \frac{2}{1/1.28 \times 10^3} \left[\frac{Z-1}{Z+1} \right] \right\}^2 + 1.432 \times 10^{-3} \left[\frac{2}{1/1.28 \times 10^3} \left[\frac{Z-1}{Z+1} \right] \right] + 1}$$

$$\therefore H(Z) = \frac{1}{6.71 \left[\frac{Z-1}{Z+1} \right]^2 + 3.67 \left[\frac{Z-1}{Z+1} \right] + 1}$$

$$\therefore H(Z) = \frac{1}{6.71 \left[\frac{1-Z^{-1}}{1+Z^{-1}} \right]^2 + 3.67 \left[\frac{1-Z^{-1}}{1+Z^{-1}} \right] + 1}$$

UEx. 2.5.15 MU - May 12, 10 Marks

For the analog transfer function

$$H(s) = \frac{3}{(s+2)(s+3)}$$

Determine $H(Z)$ with sampling period $T = 0.1$ sec using

- Impulse Invariant method
- Bilinear Transformation method

☒ **Soln. :**

$$(i) \text{ Given : } H(s) = \frac{3}{(s+2)(s+3)}$$

In P.F.E. form we can write,

$$H(s) = \frac{A_1}{s+2} + \frac{A_2}{s+3} \quad \dots (1)$$

$$\text{Here } A_1 = (s+2) \cdot \frac{3}{(s+2)(s+3)} \Big|_{s=-2}$$

$$\therefore A_1 = \frac{3}{-2+3} \quad \therefore A_1 = 3$$

$$\text{and } A_2 = (s+3) \cdot \frac{3}{(s+2)(s+3)} \Big|_{s=-3}$$

$$\therefore A_2 = \frac{3}{-3+2} \quad \therefore A_2 = -3$$

Putting these values in Equation (1) we get,

$$H(s) = \frac{3}{s+2} \cdot \frac{3}{s+3} \quad \dots(2)$$

We have transformation equation

$$P_1 = z^{0.1}$$

$$\therefore -2 \rightarrow z^{-0.1} (0.1) = 0.8187$$

and

$$-3 \rightarrow z^{-0.3} (0.1) = 0.7408$$

The $H(Z)$ is,

$$H(Z) = \frac{A_1 Z}{Z - P_1} + \frac{A_2 Z}{Z - P_2}$$

$$\therefore H(Z) = \frac{3Z}{Z - 0.8187} + \frac{3Z}{Z - 0.7408}$$

Given: $H(s) = \frac{3}{(s+2)(s+3)}$

$$\therefore H(s) = \frac{3}{s^2 + 5s + 6}$$

In BLT, the transformation equation is,

$$s \rightarrow \frac{2}{T_s} \left[\frac{Z-1}{Z+1} \right]$$

$$\therefore H(Z) = \left[\frac{3}{\left(\frac{2}{0.1}\right) \left(\frac{Z-1}{Z+1}\right) + 5} + \frac{3}{\left(\frac{2}{0.1}\right) \left(\frac{Z-1}{Z+1}\right) + 6} \right]$$

$$\therefore H(Z) = \frac{3}{400 \left(\frac{Z-1}{Z+1}\right) + 100 \left(\frac{Z-1}{Z+1}\right) + 6}$$

2.6 DESIGNING USING BUTTERWORTH APPROXIMATION

Design equations and design steps

- Let,
- A_p = Attenuation in pass band,
 - A_s = Attenuation in stop band,
 - Ω_p = Pass band edge frequency,
 - Ω_c = Cut-off frequency,
 - Ω_s = Stop band edge frequency.

In the problem the specifications of required digital filter will be given and it will be asked to design a particular discrete time butterworth filter. Then the following steps should be used:

► **Step I:** From the given specifications of digital filter equivalent analog filter as follows:

(a) For impulse invariance method

$$\Omega = \frac{\omega}{T_s}$$

(b) For bilinear transformation method

$$\Omega = \frac{2}{T_s} \tan \frac{\omega}{2}$$

Here, Ω = Frequency of analog filter

ω = Frequency of digital filter

T_s = Sampling time

► **Step II:** Calculate the order 'N' of filter using the eqn

$$N = \frac{1}{2} \times \frac{\log \left[\frac{\left(\frac{1}{2} - 1 \right)}{\left(\frac{1}{2} - 1 \right)} \right]}{\log \left(\frac{\Omega_s}{\Omega_p} \right)}$$

$$\text{or } N = \frac{\log \left[\left(\frac{1}{\delta_s^2} \right) - 1 \right]}{2 \log \left(\Omega_s / \Omega_c \right)}$$

Here δ_s = Attenuation in stop band

If the specifications are given in decibels (dB) equation,

$$N = \frac{1}{2} \frac{\log \left[\frac{10^{0.1 A_s (\text{dB})} - 1}{10^{0.1 A_p (\text{dB})} - 1} \right]}{\log \left(\frac{\Omega_s}{\Omega_p} \right)}$$

► **Step III:** Calculation of cut-off frequency (Ω_c):

The cut-off frequency (Ω_c) of analog filter is calculated

(a) For impulse invariance method

$$\Omega_c = \frac{\omega_c}{T_s}$$

(b) For bilinear transformation method

$$\Omega_c = \frac{2}{T_s} \tan \frac{\omega_c}{2}$$

When ω_c is not given then use the equation,

$$(i) \quad \Omega_c = \frac{\Omega_p}{\left(\frac{1}{A_p} - 1 \right)^{1/2N}} \text{ and if specifications are in dB}$$

Putting these values in Equation (1) we get,

$$H(s) = \frac{3}{s+2} - \frac{3}{s+3} \quad \dots(2)$$

We have transformation equation

$$P_k = e^{P_k T_s}$$

$$\therefore -2 \rightarrow e^{-2 \times 0.1} = 0.8187$$

$$\text{and} \quad -3 \rightarrow e^{-3 \times 0.1} = 0.7408$$

The $H(Z)$ is,

$$H(Z) = \frac{A_1 Z}{Z - P_1} + \frac{A_2 Z}{Z - P_2}$$

$$\therefore H(Z) = \frac{3Z}{Z - 0.8187} - \frac{3Z}{Z - 0.7408}$$

(ii) Given : $H(s) = \frac{3}{(s+2)(s+3)}$

$$\therefore H(s) = \frac{3}{s^2 + 5s + 6}$$

In BLT, the transformation equation is,

$$s \rightarrow \frac{2}{T_s} \left[\frac{Z-1}{Z+1} \right]$$

$$\therefore H(Z) = \left[\frac{\frac{3}{\left(\frac{2}{0.1}\right)^2 \left(\frac{Z-1}{Z+1}\right)^2 + 5 \left(\frac{2}{0.1}\right) \left(\frac{Z-1}{Z+1}\right) + 6} \right]$$

$$\therefore H(Z) = \frac{3}{400 \left(\frac{Z-1}{Z+1}\right)^2 + 100 \left(\frac{Z-1}{Z+1}\right) + 6}$$

2.6 DESIGNING USING BUTTERWORTH APPROXIMATION

Design equations and design steps

Let, A_p = Attenuation in pass band.

A_s = Attenuation in stop band.

Ω_p = Pass band edge frequency.

Ω_c = Cut-off frequency.

Ω_s = Stop band edge frequency.

In the problem the specifications of required digital filter will be given and it will be asked to design a particular discrete time butterworth filter. Then the following steps should be used :

► **Step I :** From the given specifications of digital filter ; obtain equivalent analog filter as follows :

(a) **For impulse invariance method**

$$\Omega = \frac{\omega}{T_s}$$

(b) **For bilinear transformation method**

$$\Omega = \frac{2}{T_s} \tan \frac{\omega}{2}$$

Here, Ω = Frequency of analog filter

ω = Frequency of digital filter

T_s = Sampling time

► **Step II :** Calculate the order 'N' of filter using the equation

$$N = \frac{1}{2} \times \frac{\log \left[\frac{\left(\frac{1}{A_s^2} - 1 \right)}{\left(\frac{1}{A_p^2} - 1 \right)} \right]}{\log \left(\frac{\Omega_s}{\Omega_p} \right)}$$

$$\text{or } N = \frac{\log \left[\left(1/\delta_s^2 \right) - 1 \right]}{2 \log \left(\Omega_s / \Omega_c \right)}$$

Here δ_s = Attenuation in stop band.

If the specifications are given in decibels (dB) then use the equation,

$$N = \frac{1}{2} \frac{\log \left[\frac{10^{0.1 A_s (\text{dB})} - 1}{10^{0.1 A_p (\text{dB})} - 1} \right]}{\log \left(\frac{\Omega_s}{\Omega_p} \right)}$$

► **Step III :** Calculation of cut-off frequency (Ω_c) :

The cut-off frequency (Ω_c) of analog filter is calculated as

(a) **For impulse invariance method**

$$\Omega_c = \frac{\omega_c}{T_s}$$

(b) **For bilinear transformation method**

$$\Omega_c = \frac{2}{T_s} \tan \frac{\omega_c}{2}$$

When ω_c is not given then use the equation,

(i) $\Omega_c = \frac{\Omega_p}{\left(\frac{1}{A_p^2} - 1 \right)^{1/2N}}$ and if specifications are in dB then

$$(ii) \Omega_c = \frac{\Omega_p}{10^{0.1 A_p - 1}}$$

► **Step IV :** Calculate the poles using,

$$P_k = \pm \Omega_c e^{j(N+2k+1)\pi/2N}, \quad k=0, 1, 2, \dots, N-1.$$

If the poles are complex conjugate then organize the poles (P_k) as complex conjugate pairs that means,

$$s_1 \text{ and } s_1^*, s_2 \text{ and } s_2^* \text{ etc.}$$

► **Step V :** Calculate the system transfer function of analog filter using,

$$H(s) = \frac{\Omega_c^N}{(s - P_1)(s - P_2) \dots}$$

and if poles are complex conjugate then,

$$H(s) = \frac{\Omega_c^N}{(s - s_1)(s - s_1^*)(s - s_2)(s - s_2^*)}$$

► **Step VI :** Design the digital filter using impulse invariance method or bilinear transformation method.

2.6.1 Solved Problems on Butterworth Approximation

Ex. 2.6.1 : digital filter has frequency specification as :

Passband frequency = $\omega_p = 0.2\pi$, Stopband frequency = $\omega_s = 0.3\pi$
 What are the corresponding specifications for pass band and stop band frequencies in analog domain if,

- (I) Impulse invariance technique is used for designing.
- (II) Bilinear transformation is used for designing.

✓ **Soln. :**

Assume sampling time $T_s = 1$.

(I) **For impulse invariance method we have,**

$$\Omega_p = \frac{\omega_p}{T_s}$$

$$\therefore \Omega_p = 0.2\pi = 0.63 \text{ rad/sec.}$$

$$\text{and } \Omega_s = \frac{\omega_s}{T_s}$$

$$\therefore \Omega_s = \frac{0.3\pi}{T_s} = 0.3\pi$$

$$\therefore \Omega_s = 0.94 \text{ rad/sec.}$$

(II) **For bilinear transformation we have,**

$$\Omega_p = \frac{2}{T_s} \tan\left(\frac{\omega_p}{2}\right)$$

$$\therefore \Omega_p = 2 \tan\left(\frac{0.2\pi}{2}\right) = 2 \tan\left(\frac{0.2 \times 180}{2}\right)$$

$$\therefore \Omega_p = 0.65 \text{ rad/sec.}$$

$$\text{and } \Omega_s = \frac{2}{T_s} \tan\left(\frac{\omega_s}{2}\right)$$

$$\therefore \Omega_s = 2 \tan\left(\frac{0.3\pi}{2}\right) = 2 \tan\left(\frac{0.3 \times 180}{2}\right)$$

$$\therefore \Omega_s = 1.019 \text{ rad/sec.}$$

UEx. 2.6.2 MU - May 08. 10 Marks

Design a second order DT butterworth filter with cut-off frequency of 1 KHz and sampling frequency of 10^4 samples/sec. by bilinear transformation.

✓ **Soln. :**

In this problem, the specifications of digital filter are not given directly. So first we have to obtain required design specifications for digital filter. Then for butterworth approximation we have to convert these specifications into specifications of equivalent analog filter. Finally using bilinear transformation we have to obtain $H(Z)$.

Given specifications of analog filter

Order of filter, $N = 2$

Cut-off frequency of analog filter, $F_c = 1 \text{ KHz} = 1000 \text{ Hz}$.

Sampling frequency, $F_s = 10^4 \text{ samples/sec.} = 10,000 \text{ Hz}$.

Part A : Calculation of the required design specification of digital filter

We have the equation to convert continuous frequency (F) into discrete frequency (f). It is,

$$f = \frac{F}{F_s} \quad \therefore f_c = \frac{F_c}{F_s}$$

$$\therefore f_c = \frac{1000}{10,000} = 0.1 \text{ cycles/sample.}$$

Now the angular frequency (frequency of digital filter) is,

$$\omega_c = 2\pi f_c = 2\pi \times 0.1$$

$$\therefore \omega_c = 0.2\pi \text{ radians/sample.}$$

Part B : Calculation of specifications of analog filter for butterworth approximation

► **Step I :** For bilinear transformation we have,

$$\Omega = \frac{2}{T_s} \tan \frac{\omega}{2} \quad \therefore \Omega_c = \frac{2}{T_s} \tan \frac{\omega_c}{2}$$

$$\text{Here, } T_s = \text{Sampling time} = \frac{1}{F_s} = \frac{1}{10,000}$$

$$\therefore \Omega_c = (2 \times 10,000) \tan\left(\frac{0.2\pi}{2}\right)$$

$$\therefore \Omega_c = 6498.39 \text{ radians/sec.}$$

The value of 'N' is given so it is not necessary to calculate it. Thus the specifications of analog filter are,

- (i) Cut-off frequency = $\Omega_c = 6498.39$ radians/sec.
- (ii) Order of filter = $N = 2$.

► **Step II :** Calculate the poles using,

$$P_k = \pm \Omega_c e^{j(N+2k+1)\pi/2N} \quad \dots k = 0, 1, \dots N-1$$

Here $N = 2$. Thus $k = 0$ to $N-1$ means $k = 0$ and $k = 1$.

(i) For $k = 0$

$$\begin{aligned} P_0 &= \pm \Omega_c e^{j(N+1)\pi/2N} = \pm 6498.39 e^{j3\pi/4} \\ &= \pm 6498.39 \left[\cos\left(\frac{3\pi}{4}\right) + j \sin\left(\frac{3\pi}{4}\right) \right] \\ &= \pm 6498.39 [-0.707 + j 0.707] \end{aligned}$$

$$\therefore P_0 = 6498.39 [-0.707 + j 0.707]$$

$$\text{and } -6498.39 [-0.707 + j 0.707]$$

$$\therefore P_0 = -4595.05 + j 4595.05 \text{ and } 4595.05 - j 4595.05$$

(ii) For $k = 1$

$$\begin{aligned} P_1 &= \pm \Omega_c e^{j(N+2+1)\pi/2N} = \pm 6498.39 e^{j5\pi/4} \\ \therefore P_1 &= \pm 6498.39 \left[\cos\left(\frac{5\pi}{4}\right) + j \sin\left(\frac{5\pi}{4}\right) \right] \\ \therefore P_1 &= \pm 6498.39 [-0.707 - j 0.707] \\ \therefore P_1 &= 6498.39 [-0.707 - j 0.707] \end{aligned}$$

$$\text{and } -6498.39 [-0.707 - j 0.707]$$

$$\therefore P_1 = -4595.05 - j 4595.05 \text{ and } 4595.05 + j 4595.05$$

Now we will plot these poles as shown in Fig. P. 2.6.2.

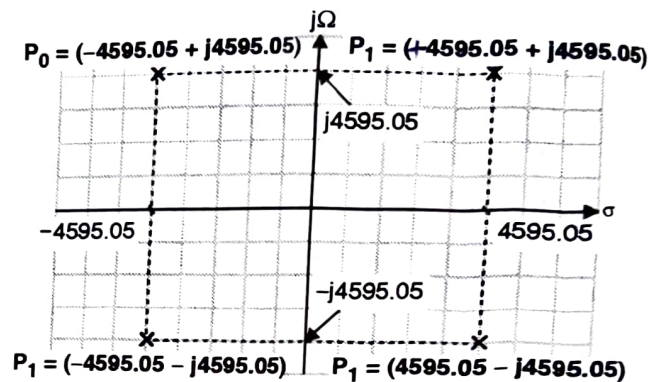


Fig. P. 2.6.2 : Plot of poles, P_k

Important Note : We know that analog filter is stable if the poles lie on the L.H.S. of s-plane. So for the filter to be stable; we have to select the poles; which are on the L.H.S. of s-plane. Observe Fig. P. 2.6.2. We will denote these poles by s_1 and s_1^* because these poles are complex conjugate of each other.

$$\therefore s_1 = -4595.05 + j 4595.05$$

$$\text{and } s_1^* = -4595.05 - j 4595.05$$

► **Step III :** The transfer function of analog filter is obtained by using the equation.

$$H(s) = \frac{\Omega_c^N}{(s - s_1)(s - s_1^*)}$$

$$\therefore H(s) = \frac{(6498.39)^2}{(s + 4595.05 - j 4595.05)(s + 4595.05 + j 4595.05)}$$

$$\therefore H(s) = \frac{(6498.39)^2}{s^2 + 4595.05s + j 4595.05s + 4595.05s + (21.11 \times 10^6) + j(21.11 \times 10^6) + j 4595.05 + j(21.11 \times 10^6) + (21.11 \times 10^6)}$$

$$\therefore H(s) = \frac{(6498.39)^2}{s^2 + 9190.1s + 42.22 \times 10^6}$$

Part C : Designing digital filter using bilinear transformation

In case of bilinear transformation $H(Z)$ is obtained by putting $s = \frac{2}{T_s} \left[\frac{Z-1}{Z+1} \right]$ in the equation of $H(s)$

Here $T_s = \frac{1}{10,000}$. Thus we have to put,

$$s = 2 \times 10^4 \left[\frac{Z-1}{Z+1} \right] \text{ in the equation of } H(s)$$

$$\therefore H(Z) = \frac{(6498.39)^2}{\left[2 \times 10^4 \left(\frac{Z-1}{Z+1} \right) \right]^2 + 183.802 \times 10^6 \left(\frac{Z-1}{Z+1} \right) + 42.22 \times 10^6}$$

This is the required transfer function for digital filter.

UEx. 2.6.3 MU - May 19, 10 Marks

Using bilinear transformation, design a butterworth filter which satisfies the following conditions :

$$0.8 \leq |H(e^{j\omega})| \leq 1 \quad 0 \leq \omega \leq 0.2\pi$$

$$|H(e^{j\omega})| \leq 0.2 \quad 0.6\pi \leq \omega \leq \pi$$

✓ **Soln. :**

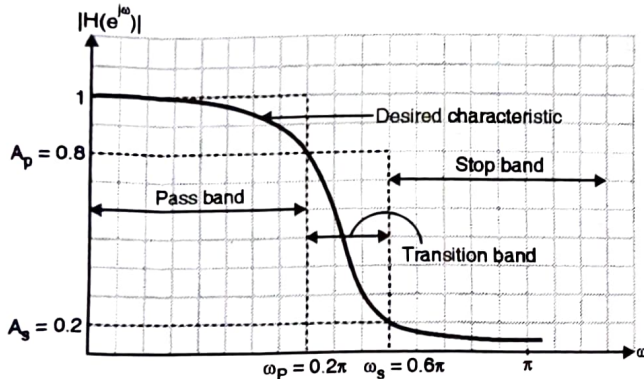


Fig. P. 2.6.3(a) : Characteristic plot for digital filter

$|H(e^{j\omega})|$ is same as $|H(\omega)|$; so the specifications are given for the required digital filter. These specifications are shown in Fig. P. 2.6.3(a).

Given specifications of digital filter are :

$$A_p = 0.8, \quad \omega_p = 0.2\pi, \quad A_s = 0.2, \quad \omega_s = 0.6\pi$$

Part A : Calculating specifications of analog filter

► **Step I :** We will obtain the values of Ω_p and Ω_s . We have to use bilinear transformation method. Thus for bilinear transformation we have,

$$\Omega = \frac{2}{T_s} \tan \frac{\omega}{2} \quad \therefore \Omega_p = \frac{2}{T_s} \tan \frac{\omega_p}{2}$$

The value of sampling time T_s is not given. So assume $T_s = 1$.

$$\therefore \Omega_p = 2 \tan \frac{0.2\pi}{2} \quad \therefore \Omega_p = 0.65$$

$$\text{Similarly, } \Omega_s = \frac{2}{T_s} \tan \frac{\omega_s}{2} = 2 \tan \frac{0.6\pi}{2}$$

$$\therefore \Omega_s = 2.75$$

Thus specifications of analog filter are :

$$A_p = 0.8, \quad \Omega_p = 0.65, \quad A_s = 0.2, \quad \Omega_s = 2.75$$

► **Step II :** Calculation of order of filter, N

It is given by,

$$N = \frac{1}{2} \times \frac{\log \left[\left(\frac{1}{A_s^2} - 1 \right) / \left(\frac{1}{A_p^2} - 1 \right) \right]}{\log \left(\frac{\Omega_s}{\Omega_p} \right)}$$

$$N = \frac{1}{2} \times \frac{\log \left[\left(\frac{1}{(0.2)^2} - 1 \right) / \left(\frac{1}{(0.8)^2} - 1 \right) \right]}{\log \left(\frac{2.75}{0.65} \right)}$$

$$= \frac{1}{2} \times \frac{\log \left[\frac{24}{0.5625} \right]}{\log \left(\frac{2.75}{0.65} \right)} = 1.3$$

Thus approximately we can take $N = 2$.

► **Step III :** Calculation of cut-off frequency, Ω_c

ω_c is not given so use the equation,

$$\Omega_c = \frac{\Omega_p}{\left(\frac{1}{A_p^2} - 1 \right)^{1/2N}} = \frac{0.65}{\left[\frac{1}{(0.8)^2} - 1 \right]^{1/4}} = 0.75$$

► **Step IV :** Calculations of the poles of $H(s)$ $P_k = \frac{\pi}{2} + (2k+1)\frac{\pi}{2}$

The poles of $H(s)$ are given by,

$$P_k = \pm \Omega_c e^{j(N+2k+1)\pi/2N}, \quad k = 0, 1, 2, \dots, N-1$$

$$\therefore P_k = \pm 0.75 e^{j(2+2k+1)\pi/4}$$

$$\therefore P_k = \pm 0.75 e^{j(3+2k)\pi/4}$$

Here range of k is from 0 to $N-1$ and $N = 2$. Thus the value of k is, $k = 0$ and $k = 1$.

For $k = 0$

$$P_0 = \pm 0.75 e^{j(3)\pi/4} = \pm 0.75 \left[\cos \left(\frac{3\pi}{4} \right) + j \sin \left(\frac{3\pi}{4} \right) \right]$$

$$\therefore P_0 = \pm 0.75 [-0.707 + j 0.707]$$

$$\therefore P_0 = \pm [-0.53 + j 0.53]$$

$$\therefore P_0 = -0.53 + j 0.53 \text{ and } 0.53 - j 0.53$$

For $k = 1$

$$P_1 = \pm 0.75 e^{j(3+2)\pi/4} = \pm 0.75 e^{j5\pi/4}$$

$$\therefore P_1 = \pm 0.75 \left[\cos \frac{5\pi}{4} + j \sin \frac{5\pi}{4} \right]$$

$$\therefore P_1 = \pm 0.75 [-0.707 - j 0.707]$$

$$\therefore P_1 = -0.53 - j 0.53 \text{ and } 0.53 + j 0.53$$

The location of poles are as shown in Fig. P. 2.6.3(b).

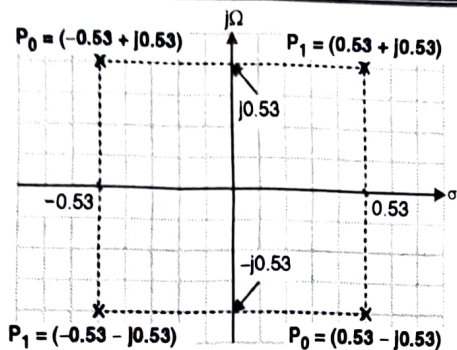


Fig. P. 2.6.3(b) : Location of poles

We know that, for the stability; the poles which are at the L.H.S. of s-plane should be selected. Thus we will select the poles $-0.53 + j 0.53$ and $-0.53 - j 0.53$. Note that these poles are complex conjugate of each other. We will denote the poles by,

$$s_1 = -0.53 + j 0.53 \text{ and } s_1^* = -0.53 - j 0.53$$

► **Step V : Calculation of $H(s)$**

The transfer function of analog filter is given by,

$$H(s) = \frac{\Omega_c^N}{(s - s_1)(s - s_1^*)}$$

$$\therefore H(s) = \frac{(0.75)^2}{(s + 0.53 - j 0.53)(s + 0.53 + j 0.53)}$$

$$\therefore H(s) = \frac{(0.75)^2}{s^2 + 0.53s + j 0.53s + 0.53s + 0.28 + j 0.28 + j 0.53s + j 0.28 + 0.28}$$

$$\therefore H(s) = \frac{(0.75)^2}{s^2 + 1.06s + 0.56}$$

$$\therefore H(s) = \frac{0.56}{s^2 + 1.06s + 0.56}$$

Part B : Obtaining transfer function for digital filter

We have to use bilinear transformation method. Then $H(Z)$

is obtained by putting $s = \frac{2}{T_s} \left(\frac{Z-1}{Z+1} \right)$ in the equation of $H(s)$.

We have assumed sampling time $T_s = 1$.

$$\therefore H(Z) = \frac{0.56}{4 \left(\frac{Z-1}{Z+1} \right)^2 + (2 \times 1.06) \left(\frac{Z-1}{Z+1} \right) + 0.56}$$

$$\therefore H(Z) = \frac{0.56}{\frac{4(Z-1)^2}{(Z+1)^2} + \frac{2.12Z - 2.12}{Z+1} + 0.56}$$

$$\therefore H(Z) = \frac{0.56(Z+1)^2}{4(Z^2 - 2Z + 1) + (2.12Z - 2.12)(Z+1) + 0.56(Z+1)^2}$$

$$= \frac{0.56(Z+1)^2}{4Z^2 - 8Z + 4 + 2.12Z^2 + 2.12Z - 2.12 + 0.56Z^2 + 1.12Z + 0.56}$$

$$\therefore H(Z) = \frac{0.56(Z+1)^2}{6.68Z^2 - 6.88Z + 2.44}$$

This is the required transfer function to design the digital filter.

(New Syllabus w.e.f academic year 21-22) (M5-70)

UEX. 2.6.4 MU - Q.5(a), Dec. 18, 10 Marks

Design a Butterworth digital IIR low pass filter using bilinear transformation by taking $T = 1$ second, to satisfy the following specification

$$0.707 \leq |H(e^{j\omega})| \leq 1.0; 0 \leq \omega \leq 0.2\pi$$

$$|H(e^{j\omega})| \leq 0.08; 0.4\pi \leq \omega \leq \pi$$

✓ **Soln. :**

Given : Specifications are as follows

$$A_p = 0.707, \omega_p = 0.2\pi, A_s = 0.08, \omega_s = 0.4\pi$$

Part A : Calculation of specifications of analog filter

► **Step I : Using BLT**

$$\Omega_p = \frac{2}{T_s} \tan\left(\frac{\omega_p}{T_s}\right) = \frac{2}{1} \tan\left(\frac{0.2\pi}{2}\right)$$

$$\therefore \Omega_p = 0.65$$

$$\Omega_s = \frac{2}{T_s} \tan\left(\frac{\omega_s}{2}\right) = \frac{2}{1} \tan\left(\frac{0.4\pi}{2}\right)$$

$$\therefore \Omega_s = 1.45$$

Thus specifications of analog filter as,

$$A_p = 0.707, \Omega_p = 0.65, A_s = 0.08, \Omega_s = 1.45$$

► **Step II : Calculation of order of filter, N**

$$N = \frac{1}{2} \times \frac{\log\left[\left(\frac{1}{A_s^2} - 1\right) / \left(\frac{1}{A_p^2} - 1\right)\right]}{\log\left(\frac{\Omega_s}{\Omega_p}\right)}$$

$$\frac{2}{T} \left(\frac{Z-1}{Z+1} \right)^N = \frac{1}{2} \times \frac{\log\left[\left(\frac{1}{0.08^2} - 1\right) / \left(\frac{1}{0.707^2} - 1\right)\right]}{\log\left(\frac{1.45}{0.65}\right)}$$

$$= \frac{1}{2} \frac{\log\left[\frac{155.25}{1.006}\right]}{\log\left(\frac{1.45}{0.65}\right)}$$

$$\therefore N = \frac{1}{2} \cdot \frac{2.1907}{0.3484} = 3.1439$$

$$\therefore N \approx 4$$

► **Step III : Calculation of cutoff frequency (Ω_c)**

$$\Omega_c = \frac{\Omega_p}{\left(\frac{1}{A_p^2} - 1\right)^{\frac{1}{2N}}} = \frac{0.65}{\left(\frac{1}{0.707^2} - 1\right)^{\frac{1}{8}}}$$

$$\therefore \Omega_c = 0.65$$

Step IV : Calculation of poles of H (s)

The poles are given by,

$$P_k = \pm \Omega_c e^{j(N+2k+1)\pi/2N} \quad \dots k = 0, 1, \dots, N-1$$

$$\therefore P_k = \pm 0.65 e^{j(4+2k+1)\frac{\pi}{8}} \quad \dots k = 0 \text{ to } 3$$

$$\therefore P_k = \pm 0.65 e^{j(5+2k)\frac{\pi}{8}} \quad \dots k = 0 \text{ to } 3$$

For k = 0

$$P_0 = \pm 0.65 e^{j(5\pi/8)} = \pm 0.65 [-0.38 + j0.92]$$

$$\therefore P_0 = -0.25 + j0.6 \text{ and } 0.25 - j0.6$$

For k = 1

$$P_1 = \pm \Omega_c e^{j(5+2)\frac{\pi}{8}} = \pm 0.65 e^{j(7\pi/8)}$$

$$= \pm 0.65 [-0.92 + j0.38]$$

$$\therefore P_1 = -0.6 + j0.25 \text{ and } 0.6 - j0.25$$

For k = 2

$$\therefore P_2 = \pm 0.65 e^{j(5+4)\frac{\pi}{8}} = \pm 0.65 e^{j(9\pi/8)}$$

$$= \pm 0.65 [-0.92 - j0.38]$$

$$\therefore P_2 = -0.6 - j0.25 \text{ and } 0.6 + j0.25$$

For k = 3

$$\therefore P_3 = \pm 0.65 e^{j(5+6)\frac{\pi}{8}} = \pm 0.65 e^{j(11\pi/8)}$$

$$= \pm 0.65 [-0.38 - j0.92]$$

$$\therefore P_3 = -0.25 - j0.6 \text{ and } 0.25 + j0.6$$

These poles are plotted as shown in Fig. P. 2.6.4.

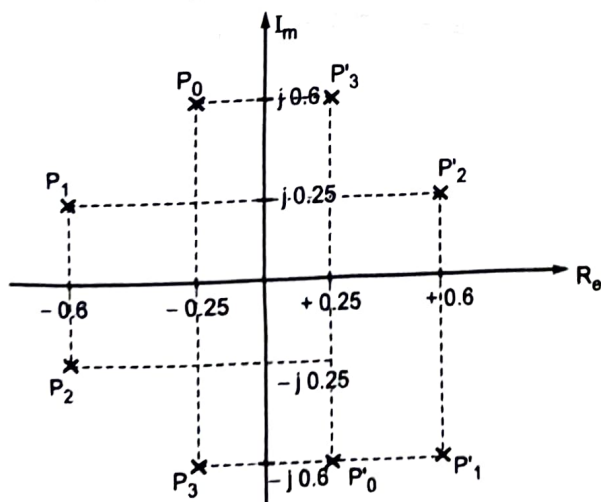


Fig. P. 2.6.4

For the stability, select the poles which are at L.H.S.

Step V : Calculation of H (s)

The transfer function of analog filter is given by,

$$H(s) = \frac{\Omega_c^N}{(s - P_0)(s - P_1)(s - P_2)(s - P_3)}$$

$$\therefore H(s) =$$

$$\frac{(0.65)^4}{(s + 0.25 - j0.6)(s + 0.6 - j0.25)(s + 0.6 + j0.25)(s + 0.25 + j0.6)}$$

$$\therefore H(s) = \frac{0.1785}{[(s + 0.25)^2 - j^2(0.6)^2][(s + 0.6)^2 - j^2(0.25)^2]}$$

$$\therefore H(s) = \frac{0.1785}{[(s + 0.25)^2 + 0.36][(s + 0.6)^2 + 0.0625]}$$

Part B : Calculation of transfer function of digital filterUsing B/T, it is obtained by replacing s by $\frac{2}{T_1} \left(\frac{Z-1}{Z+1} \right)$.Here $T_1 = 1$

$$\therefore H(s) =$$

$$\frac{0.1785}{\left[\left(2 \left(\frac{Z-1}{Z+1} \right) + 0.25 \right)^2 + 0.36 \right] \left[\left(2 \left(\frac{Z-1}{Z+1} \right) + 0.6 \right)^2 + 0.0625 \right]}$$

This is the transfer function of digital filter.

UEx. 2.6.5 MU - May 06, 10 Marks

(a) Find the order and cut-off frequency of a digital filter with following specifications :

$$0.89 \leq |H(e^{j\omega})| \leq 1, \quad 0 \leq \omega \leq 0.4\pi$$

$$|H(e^{j\omega})| \leq 0.18, \quad 0.6\pi \leq \omega \leq \pi$$

Use impulse invariance method.

(b) Draw its poles.

✓ Soln. :

(a) The given specifications are :

$$0.89 \leq |H(e^{j\omega})| \leq 1, \quad 0 \leq \omega \leq 0.4\pi$$

$$|H(e^{j\omega})| \leq 0.18, \quad 0.6\pi \leq \omega \leq \pi$$

$$\text{Thus, } A_p = 0.89, \quad \omega_p = 0.4\pi$$

$$A_s = 0.18, \quad \omega_s = 0.6\pi$$

Calculation of order of filter

Order of filter is given by,

$$N = \frac{1}{2} \frac{\log \left[\frac{\left(\frac{1}{A_s} \right)^2 - 1}{\left(\frac{1}{A_p} \right)^2 - 1} \right]}{\log \left(\frac{\Omega_s}{\Omega_p} \right)} \quad \dots (1)$$

First we will obtain values of Ω_s and Ω_p .

For impulse invariance method,

$$\Omega_p = \frac{\omega_p}{T_s} \text{ and } \Omega_s = \frac{\omega_s}{T_s}$$

Here sampling time T_s is not given. So assume $T_s = 1$ sec.

$$\therefore \Omega_p = \omega_p = 0.4\pi \text{ and } \Omega_s = \omega_s = 0.6\pi$$

Putting these values in Equation (1) we get,

$$N = \frac{1}{2} \times \frac{\log \left[\frac{1}{(0.18)^2 - 1} \right]}{\log \left(\frac{0.6\pi}{0.4\pi} \right)}$$

$$= \frac{1}{2} \times \frac{2.056}{0.176} = 5.83$$

$$\text{Thus order of filter} = N \approx 6$$

Calculation of cut-off frequency

The cut-off frequency is given by,

$$\Omega_c = \frac{\Omega_p}{\left(\frac{1}{A_p^2} - 1 \right)^{\frac{1}{2N}}} = \frac{0.4\pi}{\left[\frac{1}{(0.89)^2 - 1} \right]^{\frac{1}{2 \times 6}}}$$

$$\therefore \Omega_c = 1.405$$

(b) The poles are calculated using,

$$P_k = \pm \Omega_c e^{j(N+2k+1)\pi/2N}, \quad k=0, 1, \dots, N-1$$

We have $N=6$. Thus k varies from 0 to 5.

For $k=0 \Rightarrow$

$$P_0 = \pm 1.405 e^{j(6+0+1)\pi/12}$$

$$= \pm 1.405 e^{j\frac{7\pi}{12}}$$

$$\therefore P_0 = \pm 1.405 \left[\cos \frac{7\pi}{12} + j \sin \frac{7\pi}{12} \right]$$

$$= \pm 1.405 [-0.26 + j 0.97]$$

$$\therefore P_0 = -0.37 + j 1.37, 0.37 - j 1.37$$

For $k=1 \Rightarrow$

$$P_1 = \pm 1.405 e^{j(6+2+1)\pi/12} = \pm 1.405 e^{j\frac{9\pi}{12}}$$

$$= \pm 1.405 \left[\cos \frac{9\pi}{12} + j \sin \frac{9\pi}{12} \right]$$

$$= \pm 1.405 [-0.707 + j 0.707]$$

$$\therefore P_1 = -1 + j1, 1 - j1$$

For $k=2 \Rightarrow$

$$P_2 = \pm 1.405 e^{j(6+4+1)\pi/12} = \pm 1.405 e^{j\frac{11\pi}{12}}$$

$$= \pm 1.405 \left[\cos \frac{11\pi}{12} + j \sin \frac{11\pi}{12} \right]$$

$$= \pm 1.405 [-0.97 + j 0.26]$$

$$\therefore P_2 = -1.36 + j 0.36, 1.36 - j 0.36$$

For $k=3 \Rightarrow$

$$P_3 = \pm 1.405 e^{j(6+6+1)\pi/12} = \pm 1.405 e^{j\frac{13\pi}{12}}$$

$$= \pm 1.405 \left[\cos \frac{13\pi}{12} + j \sin \frac{13\pi}{12} \right]$$

$$= \pm 1.405 [-0.97 - j 0.26]$$

$$\therefore P_3 = -1.36 - j 0.36, 1.36 + j 0.36$$

For $k=4 \Rightarrow$

$$P_4 = \pm 1.405 e^{j(6+8+1)\pi/12} = \pm 1.405 e^{j\frac{15\pi}{12}}$$

$$= \pm 1.405 \left[\cos \frac{15\pi}{12} + j \sin \frac{15\pi}{12} \right]$$

$$= \pm 1.405 [-1 - j 0.707]$$

$$\therefore P_4 = -1 - j1, 1 + j1$$

For $k=5 \Rightarrow$

$$P_5 = \pm 1.405 e^{j(6+10+1)\pi/12} = \pm 1.405 e^{j\frac{17\pi}{12}}$$

$$= \pm 1.405 \left[\cos \frac{17\pi}{12} + j \sin \frac{17\pi}{12} \right]$$

$$= \pm 1.405 [-0.26 - j 0.97]$$

$$\therefore P_5 = -0.37 - j 1.37, 0.37 + j 1.37$$

The plot of poles is shown in Fig. P. 2.6.4.

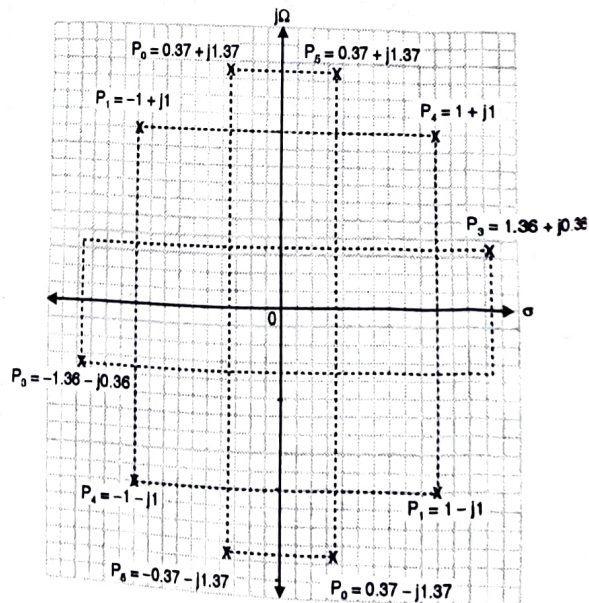


Fig. P. 2.6.5

Ex. 2.6.6 : A digital low pass IIR filter is to be designed with butter-worth approximation using bilinear transformation technique. Find the order of filter to meet the following specifications :

- Passband magnitude is constant within 1 dB for frequencies below 0.2π .
- Stopband attenuation is greater than 15 dB for frequencies between 0.3π to π .

✓ **Soln. :**

The given specifications for the desired digital IIR filter are :

$$A_p = 1 \text{ dB}, \omega_p = 0.2\pi \text{ and } A_s = 15 \text{ dB}, \omega_s = 0.3\pi$$

Part A : Calculation of specifications for analog filter

For the bilinear transformation we have,

$$\Omega = \frac{2}{T_s} \tan \frac{\omega}{2}$$

The values of T_s is not given, so assume $T_s = 1$.

$$\therefore \Omega = 2 \tan \frac{\omega}{2}$$

Thus we can write,

$$\Omega_p = 2 \tan \frac{\omega_p}{2} = 2 \tan \frac{0.2\pi}{2}$$

$$\therefore \Omega_p = 0.65$$

$$\text{Similarly, } \Omega_s = 2 \tan \frac{\omega_s}{2} = 2 \tan \frac{0.3\pi}{2}$$

$$\therefore \Omega_s = 1.02$$

Thus specifications of analog filter are :

$$A_p = 1 \text{ dB}, \Omega_p = 0.65$$

$$\text{and } A_s = 15 \text{ dB}, \Omega_s = 1.02$$

Part B : Calculation of the order of filter (N)

The specifications are given in dB; so we have to use the formula,

$$N = \frac{1}{2} \frac{\log \left[\frac{10^{0.1 A_s \text{ dB}} - 1}{10^{0.1 A_p \text{ dB}} - 1} \right]}{\log \left(\frac{\Omega_s}{\Omega_p} \right)}$$

$$\therefore N = \frac{1}{2} \frac{\log \left[\frac{10^{(0.1 \times 15)} - 1}{10^{(0.1 \times 1)} - 1} \right]}{\log \left(\frac{1.02}{0.65} \right)}$$

$$= \frac{1}{2} \times \frac{2.073}{0.1957}$$

$$\therefore N = 5.296$$

Thus the order of filter is $N = 6$.

Ex. 2.6.7 : An IIR low pass filter is required to meet the following specifications :

Passband peak to peak ripple : $\leq 1 \text{ dB}$

Passband edge : 1.2 kHz

Stopband attenuation : $\geq 40 \text{ dB}$

Stopband edge : 2.5 kHz

Sample rate : 8 kHz

The filter is to be designed by performing BLT on an analog system function of required order butterworth filter so as to meet the specifications in the implementation.

✓ **Soln. :**

Given specifications are,

Attenuation in passband (peak to peak ripple) = $A_p = 1 \text{ dB}$

Attenuation in stopband = $A_s = 40 \text{ dB}$

Passband edge frequency = $F_{PB} = 1.2 \text{ kHz}$

Stopband edge frequency = $F_{SB} = 2.5 \text{ kHz}$

Sampling frequency = $F_s = 8 \text{ kHz}$

Note : Since every frequency is in kHz it is not required to convert into Hz.

Part A : Calculation of the required design specifications of digital filter

$$\text{We have, } f_{PB} = \frac{F_{PB}}{F_s} = \frac{1.2 \text{ kHz}}{8 \text{ kHz}} = 0.15$$

$$\therefore \omega_p = 2\pi f_{PB} = 2\pi \times 0.15 = 0.94$$

$$\text{Similarly } f_{SB} = \frac{F_{SB}}{F_s} = \frac{2.5 \text{ kHz}}{8 \text{ kHz}} = 0.31$$

$$\therefore \omega_s = 2\pi f_{SB} = 2\pi \times 0.31 = 1.96$$

Thus specifications of digital filter are,

$$A_p = 1 \text{ dB}, \quad \omega_p = 0.94$$

$$A_s = 40 \text{ dB}, \quad \omega_s = 1.96$$

Part B : Calculation of specifications of analog filter for butterworth approximation

► **Step I :** For BLT we have,

$$\Omega = \frac{2}{T_s} \tan \frac{\omega}{2}$$

$$\therefore \Omega_p = \frac{2}{T_s} \tan \frac{\omega_p}{2}$$

$$\text{Here } T_s = \frac{1}{F_s} = \frac{1}{8}$$

$$\therefore \Omega_p = 16 \tan \frac{0.94}{2} \quad \therefore \Omega_p = 8.13$$

$$\text{and } \Omega_s = \frac{2}{T_s} \tan \frac{\omega_s}{2}$$

$$\therefore \Omega_s = (2 \times 8) \tan \frac{1.96}{2} \quad \therefore \Omega_s = 23.85$$

Thus specification of analog filter are,

$$A_p = 1 \text{ dB}, \quad \Omega_p = 8.13$$

$$A_s = 40 \text{ dB}, \quad \Omega_s = 23.85$$

► Step II : Calculation of order of filter

The order of filter is given by,

$$N = \frac{1}{2} \frac{\log \left[\frac{10^{0.1A_s} - 1}{10^{0.1A_p} - 1} \right]}{\log \left(\frac{\Omega_s}{\Omega_p} \right)}$$

$$\therefore N = \frac{1}{2} \frac{\log \left[\frac{10^{0.1(40)} - 1}{10^{0.1(1)} - 1} \right]}{\log \left(\frac{23.85}{8.13} \right)} = \frac{1}{2} \times \frac{4.586}{0.467}$$

$$\therefore N = 4.9 \approx 5$$

► Step III : Calculation of cut-off frequency

The cut-off frequency (Ω_c) is,

$$\Omega_c = \frac{\Omega_p}{10^{0.1A_p - 1}}$$

$$\Omega_c = \frac{8.13}{10^{0.1(1)} - 1}$$

$$\therefore \Omega_c = 31.4$$

► Step IV : Calculation of poles of $H(s)$

The poles of $H(s)$ are given by,

$$P_k = \pm \Omega_c e^{j(N+2k+1)\pi/2N}$$

$$\text{Here } N = 5$$

$$\dots k = 0, 1, 2, \dots N-1$$

$$\therefore P_k = \pm 31.4 e^{j(6+2k)\pi/10}$$

$$\dots k = 0, 1, 2, 3, 4$$

$$\text{For } k=0$$

$$P_0 = \pm 31.4 e^{j\frac{6\pi}{10}} = \pm 31.4 e^{j1.88}$$

$$\therefore P_0 = \pm 31.4 [\cos 1.88 + j \sin 1.88]$$

(New Syllabus w.e.f academic year 21-22) (M5-70)

$$= \pm 31.4 [-0.3 + j 0.95]$$

$$\therefore P_0 = -9.42 + j 29.83 \text{ and } 9.42 - j 29.83$$

$$\text{For } k=1 \quad P_1 = \pm 31.4 e^{j(5+2+1)\pi/10}$$

$$= \pm 31.4 e^{j\frac{8\pi}{10}}$$

$$\therefore P_1 = \pm 31.4 \left[\cos \frac{8\pi}{10} + j \sin \frac{8\pi}{10} \right]$$

$$= \pm 31.4 [-0.8 + j 0.59]$$

$$\therefore P_1 = -25.12 + j 18.5 \text{ and } 25.12 - j 18.5$$

$$\text{For } k=2 \quad P_2 = \pm 31.4 e^{j(5+4+1)\pi/10} = \pm 31.4 e^{j\pi}$$

$$\therefore P_2 = \pm 31.4 [\cos \pi + j \sin \pi]$$

$$= \pm 31.4 [-1 + j 0] = \pm 31.4 [-1]$$

$$\therefore P_2 = 31.4 + j 0 \text{ and } -31.4 + j 0$$

$$\text{For } k=3 \quad P_3 = \pm 31.4 e^{j(5+6+1)\pi/10}$$

$$= \pm 31.4 e^{j\frac{12\pi}{10}}$$

$$\therefore P_3 = \pm 31.4 \left[\cos \left(\frac{12\pi}{10} \right) + j \sin \left(\frac{12\pi}{10} \right) \right]$$

$$= \pm 31.4 [-0.8 - j 0.59]$$

$$\therefore P_3 = -25.12 - j 18.52 \text{ and } 25.12 + j 18.52$$

$$\text{For } k=4 \quad P_4 = \pm 31.4 e^{j(5+8+1)\pi/10}$$

$$= \pm 31.4 e^{j\frac{14\pi}{10}}$$

$$\therefore P_4 = \pm 31.4 \left[\cos \frac{14\pi}{10} + j \sin \left(\frac{14\pi}{10} \right) \right]$$

$$= \pm 31.4 [-0.3 - j 0.95]$$

$$\therefore P_4 = -9.42 - j 29.83 \text{ and } 9.42 + j 29.83$$

These poles are plotted as shown in Fig. P. 2.6.7.

We know that for the stability of filter; we have to select the poles which are on the L.H.S. of s -plane. So we will select the following poles:

$$s_1 = -9.42 + j 29.83; \quad s_1^* = -9.42 - j 29.83$$

$$s_2 = -25.12 + j 18.5; \quad s_2^* = -25.12 - j 18.5$$

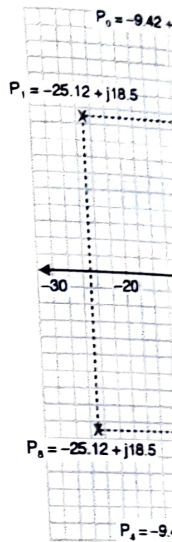
and

$$s_3 = -31.4 + j 0$$

► Step V : Calculation of $H(s)$

The transfer function of analog filter is given by,

$$H(s) = \frac{\Omega_c^N}{(s-s_1)(s-s_1^*)(s-s_2)(s-s_2^*)(s-s_3)}$$



$$\therefore H(s) = \frac{1}{(s+9.42-j29.83)(s+9.42+j29.83)(s^2+18.84s+978.5)}$$

$$\therefore H(s) = \frac{1}{(s^2+18.84s+978.5)(s^2+31.4s+978.5)}$$

Part C : Obtaining transfer function for digital filter

The transfer function of digital filter is obtained

$$s = \frac{2}{T_s} \left[\frac{Z-1}{Z+1} \right]$$

$$\therefore s = 16 \left[\frac{Z-1}{Z+1} \right]$$

Thus we get,

$$H(Z) = \frac{1}{256 \left(\frac{Z-1}{Z+1} \right)^2 + 301.5 \left(\frac{Z-1}{Z+1} \right) + 978.5}$$

Ex. 2.6.8 : Design a third order Butterworth digital filter

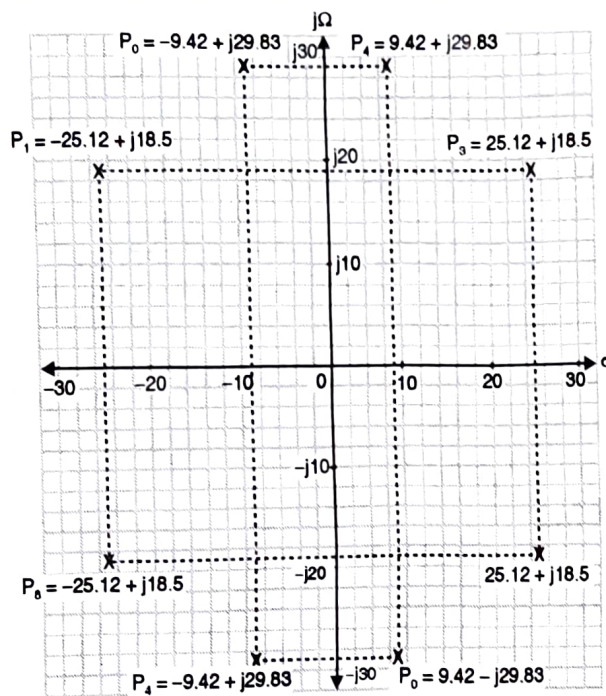
✓ Soln. : From the table of butterworth polynomials

$$H(s) = \frac{1}{(s+1)(s^2+s+1)}$$

$$\therefore H(s) = \frac{1}{(s+1)(s+0.5-j0.87)(s+0.5+j0.87)}$$

Thus poles of analog filter are,

(New Syllabus w.e.f academic year 21-22)

Fig. P. 2.6.7 : Poles of $H(s)$

$$\therefore H(s) = \frac{(31.4)^5}{(s + 9.42 - j29.83)(s + 9.42 + j29.83)(s + 25.12 - j18.5)(s + 25.12 + j18.5)(s + 31.4)}$$

$$\therefore H(s) = \frac{(31.4)^5}{(s^2 + 18.84s + 978.5)(s^2 + 50.24s + 973.26)(s + 31.4)}$$

Part C : Obtaining transfer function for digital filter

The transfer function of digital filter is obtained by putting,

$$s = \frac{2}{T_s} \left[\frac{Z-1}{Z+1} \right] \quad \text{Here } T_s = \frac{1}{F_s} = \frac{1}{8}$$

$$\therefore s = 16 \left[\frac{Z-1}{Z+1} \right]$$

Thus we get,

$$H(Z) = \frac{(31.4)^5}{\left[256 \left(\frac{Z-1}{Z+1} \right)^2 + 301.5 \left(\frac{Z-1}{Z+1} \right) + 978.5 \right] \left[256 \left(\frac{Z-1}{Z+1} \right)^2 + 803.84 \left(\frac{Z-1}{Z+1} \right) + 973.26 \right] \left[16 \left(\frac{Z-1}{Z+1} \right) + 31.4 \right]}$$

Ex. 2.6.8 : Design a third order Butterworth digital filter using impulse invariance method. Assume sampling period $T = 1$ sec.

✓ **Soln. :** From the table of butterworth polynomials, for order 3,

$$H(s) = \frac{1}{(s+1)(s^2+s+1)}$$

$$\therefore H(s) = \frac{1}{(s+1)(s+0.5-j0.87)(s+0.5+j0.87)} \quad \dots(1)$$

Thus poles of analog filter are,

$$P_1 = -1, P_2 = -0.5 + j0.87, P_3 = -0.5 - j0.87$$

In P.F.E. form Equation (1) can be written as,

$$H(s) = \frac{A_1}{s+1} + \frac{A_2}{s-(-0.5+j0.87)} + \frac{A_3}{s-(-0.5-j0.87)}$$

$$A_1 = (s+1) \cdot \frac{1}{(s+1)[s-(-0.5+j0.87)][s-(-0.5-j0.87)]} \Big|_{s=-1}$$

$$\therefore A_1 = \frac{1}{[-1-(-0.5+j0.87)][-1-(-0.5-j0.87)]} = \frac{1}{(-0.5-j0.87)(-0.5+j0.87)}$$

$$\therefore A_1 = 1$$

$$A_2 = (s-(-0.5+j0.87)) \cdot \frac{1}{(s+1)[s-(-0.5+j0.87)][s-(-0.5-j0.87)]} \Big|_{s=-0.5+j0.87}$$

$$\therefore A_2 = \frac{1}{(-0.5+j0.87+1)[-0.5+j0.87+0.5+j0.87]}$$

$$\therefore A_2 = \frac{1}{(0.5+j0.87)(j1.74)} = -1.5138 + j0.87$$

$$\therefore A_2 = -0.49 - j0.29$$

Now A_3 is complex conjugate of A_2 .

$$\therefore A_3 = -0.49 + j0.29$$

Putting these values in Equation (2) we get,

$$H(s) = \frac{1}{s+1} + \frac{(-0.49-j0.29)}{s-(-0.5+j0.87)} + \frac{(-0.49+j0.29)}{s-(-0.5-j0.87)}$$

In impulse invariance, the transformation equation is,

$$\frac{1}{s-P_k} \rightarrow \frac{1}{1-e^{P_k T_s} \cdot Z^{-1}}$$

$$\therefore \frac{1}{s+1} \rightarrow \frac{1}{1-e^{-1} \cdot Z^{-1}} = \frac{1}{1-0.368Z^{-1}}$$

$$\frac{1}{s-(-0.5+j0.87)} \rightarrow \frac{1}{1-e^{(-0.5+j0.87)T_s} \cdot Z^{-1}} = \frac{1}{1-0.6e^{j0.87} \cdot Z^{-1}}$$

$$\frac{1}{s-(-0.5-j0.87)} \rightarrow \frac{1}{1-e^{(-0.5-j0.87)T_s} \cdot Z^{-1}} = \frac{1}{1-0.6e^{-j0.87} \cdot Z^{-1}}$$

$$\therefore H(Z) = \frac{1}{1-0.368Z^{-1}} + \frac{(-0.49-j0.29)}{1-0.6e^{j0.87} \cdot Z^{-1}} + \frac{(-0.49+j0.29)}{1-0.6e^{-j0.87} \cdot Z^{-1}}$$

$$\therefore H(Z) = \frac{Z}{Z-0.368} + \frac{(-0.49-j0.29)Z}{Z-0.6e^{j0.87}} + \frac{(-0.49+j0.29)Z}{Z-0.6e^{-j0.87}}$$

UEx. 2.6.9 MU - Dec. 12, 10 Marks

Obtain the analog transfer function of a Butterworth low-pass filter with following specifications :

Passband edge (Ω_p) = 250 rad/sec.

Passband attenuation ≤ 0.1 dB

Stopband edge (Ω_s) = 2000 rad/sec

Stopband attenuation ≥ 60 dB

☒ Soln. :

► Step I : Calculation of order

$$N = \frac{1}{2} \frac{\log \left[\frac{10^{0.1 A_s} - 1}{10^{0.1 A_p} - 1} \right]}{\log \left[\frac{\Omega_s}{\Omega_p} \right]}$$

$$N = \frac{1}{2} \frac{\log \left[\frac{10^{0.1 \times 60} - 1}{10^{0.1 \times 0.1} - 1} \right]}{\log \left[\frac{2000}{250} \right]}$$

$$\therefore N = 4.225 \approx 5$$

► Step II : Calculation of cut-off frequency

$$\Omega_c = \frac{\Omega_p}{10^{0.1 A_p} - 1} = \frac{250}{10^{0.1 \times 0.1} - 1}$$

$$\therefore \Omega_c = 10732.84$$

Step III : Calculation of poles

$$P_k = \pm \Omega_c e^{j(N+2k+1)\pi/2N} \quad \dots k = 0 \text{ to } N-1$$

$$\therefore P_k = \pm 10732.84 e^{j(5+2k+1)\frac{\pi}{10}} \quad \dots k = 0 \text{ to } 4$$

For $k = 0$

$$P_0 = \pm 10732.84 e^{j\frac{6\pi}{10}} = \pm 10732.84 [-0.309 + j0.951]$$

$$= \pm [-3316.62 + j 10206.93]$$

$$P_0 = -3316.62 + j 10206.93 \text{ and}$$

$$P'_0 = 3316.62 - j 10206.93$$

For $k = 1$

$$P_1 = \pm 10732.84 e^{j\frac{8\pi}{10}}$$

$$= \pm 10732.84 [-0.809 + j0.5877]$$

$$= \pm [-8682.86 + j 6308.6]$$

$$\therefore P_1 = -8682.82 + j 6308.6, P'_1 = 8682.82 - j 6308.6$$

For $k = 2$

$$P_2 = \pm 10732.84 e^{j\pi}$$

$$= \pm 10732.84 [-1] = \pm [-10732.84]$$

$$\therefore P_2 = -10732.84, P'_2 = 10732.84$$

For $k = 3$

$$P_3 = \pm 10732.84 e^{j\frac{12\pi}{10}}$$

$$= \pm 10732.84 [-0.809 - j 0.5877]$$

$$= \pm [-8682.82 - j 0.5877]$$

$$\therefore P_3 = -8682.82 - j6308.6, P'_3 = 8682.82 + j6308.6$$

For $k = 4$

$$P_4 = \pm 10732.84 e^{j\left(\frac{14\pi}{10}\right)}$$

$$= \pm 10732.84 [-0.309 - j 0.951]$$

$$\therefore P_4 = \pm [-3316.62 - j 10206.93]$$

$$\therefore P_4 = -3316.62 - j 10206.93$$

$$\text{and } P'_4 = 3316.62 + j 10206.93$$

These poles are plotted as shown in Fig. P. 2.6.9.

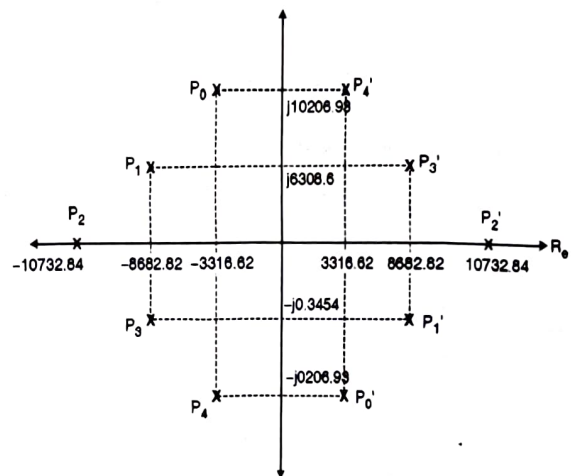


Fig. P. 2.6.9

Step IV : For the stability we will select the poles at L.H.S. only

$$\text{Thus, } H(s) = \frac{\Omega_c^N}{(s - P_0)(s - P_1)(s - P_2)(s - P_3)(s - P_4)}$$

$$\therefore H(s) = \frac{(10732.84)^5}{(s + 3316.62 - j10206.93)(s + 8682.82 - j6308.6)(s + 10732.84)(s + 8682.82 + j6308.6)(s + 3316.62 + j10206.93)}$$

This is analog transfer function of the filter.

UEx. 2.6.10 MU - Dec. 15, 10 Marks

Design a digital low pass IIR Butterworth filter for the following specification :

Passband ripple : $\leq 1\text{ dB}$;

Passband edge : 4 KHz

Stopband attenuation : 40 dB ;

Stop edge : 8 KHz

Sampling Rate : 24 KHz

Use bilinear transformation.

Soln. :

Given, $A_p = 1\text{ dB}$, $A_s = 40\text{ dB}$, $F_{PB} = 4\text{ kHz}$, $F_{SB} = 8\text{ kHz}$,
 $F_s = 24\text{ kHz}$

Part A : Calculation of the required design specification of digital filter

$$F_{PB} = \frac{F_{PB}}{F_s} = \frac{4}{24} = 0.167$$

$$\therefore \omega_p = 2\pi F_{PB} = 1.047$$

$$\text{and } F_{SB} = \frac{F_{SB}}{F_s} = \frac{8}{24} = 0.33$$

$$\therefore \omega_s = 2.094$$

Part B : Specifications of analog filter for butterworth approximation

► **Step I :** $\Omega_p = \frac{2}{T_s} \tan \frac{\omega_p}{2}$

Here $T_s = \frac{1}{F_s} = \frac{1}{24} = 0.0417$

$\therefore \Omega_p = \frac{2}{0.0417} \tan \frac{1.047}{2} = 27.68$

and $\Omega_s = \frac{2}{T_s} \tan \frac{\omega_s}{2} = \frac{2}{0.0417} \tan \frac{2.094}{2} = 83.034$

► **Step II :** Order of filter is,

$$N = \frac{1}{2} \cdot \frac{\log \left[\frac{10^{0.1 A_s} - 1}{10^{0.1 A_p} - 1} \right]}{\log \left(\frac{\Omega_s}{\Omega_p} \right)} = \frac{1}{2} \cdot \frac{\log \left[\frac{10^{0.1 (40)} - 1}{10^{0.1 (1)} - 1} \right]}{\log \left(\frac{83.034}{27.68} \right)}$$

$\therefore N = \frac{1}{2} \cdot \frac{4.59}{0.48} = 4.78 \approx 5$

► **Step III :** The cutoff frequency is given by,

$$\Omega_c = \frac{\Omega_p}{10^{0.1 A_p} - 1} = \frac{27.68}{10^{0.1 \times 1} - 1} = 106.9$$

► **Step IV :** Calculation of poles

The poles are given by,

These poles are shown in Fig. P. 2.6.10.

For stability, we will consider the poles at L.H.S. only.

$\therefore P_0 = -33.03 + j101.55j,$

$P_1 = -86.48 + j63.07, P_2 = -106.9$

$P_3 = -86.48 - j63.07, P_4 = -33.03 - j101.55j$

► **Step V :** The transfer function is given by,

$$\begin{aligned} H(s) &= \frac{\Omega_s^N}{(s - P_0)(s - P_1)(s - P_2)(s - P_3)(s - P_4)} \\ &= \frac{(83.034)^5}{(s + 33.03 - j101.55)(s + 86.48 - j63.07)(s + 106.9)(s + 86.48 + j63.07)(s + 33.03 + j101.55)} \\ &= \frac{(83.034)^5}{[(s + 33.03)^2 - j^2 (101.55)^2][(s + 86.48)^2 - j^2 (63.07)^2](s + 106.9)} \\ H(s) &= \frac{(83.034)^5}{[s^2 + 66.06s + 11403.38][s^2 + 172.96s + 11456.6](s + 106.9)} \end{aligned}$$

Part C : Transfer function for digital filter is obtained by using BLT

$$s = \frac{2}{T_s} \left[\frac{Z-1}{Z+1} \right] = \frac{2}{0.0417} \left[\frac{Z-1}{Z+1} \right] = 47.96 \left[\frac{Z-1}{Z+1} \right]$$

$$H(Z) = \frac{(83.034)^5}{\left\{ (47.96)^2 \left(\frac{Z-1}{Z+1} \right)^2 + 66.06 \times 47.96 \left(\frac{Z-1}{Z+1} \right) + 11403.38 \right\} \left\{ (47.96)^2 \left(\frac{Z-1}{Z+1} \right)^2 + 172.96 \times 47.96 \left(\frac{Z-1}{Z+1} \right) + 11456.6 \right\} \left[47.96 \left(\frac{Z-1}{Z+1} \right) + 106.9 \right]}$$

$$\begin{aligned} P_k &= \pm \Omega_c e^{j(N+2k+1)\pi/2N} \dots k=0 \text{ to } N-1 \\ \therefore P_k &= \pm 106.9 e^{j(5+2k+1)\pi/2 \times 5} \dots k=0 \text{ to } 4 \\ \therefore P_k &= \pm 106.9 e^{j(6+2k)\pi/10} \dots k=0 \text{ to } 4 \end{aligned}$$

For $k=0 \therefore P_0 = \pm 106.9 e^{j\frac{6\pi}{10}} = \pm 106.9 [-0.309 + j0.95]$
 $= \pm (-33.03 + j101.55)$

For $k=1 \therefore P_1 = \pm 106.9 e^{j\frac{8\pi}{10}} = \pm 106.9 [-0.809 + j0.59]$
 $= \pm (-86.48 + j63.07)$

For $k=2 \therefore P_2 = \pm 106.9 e^{j\pi} = \pm 106.9 [-1 + j0] = \pm 106.9$

For $k=3 \therefore P_3 = \pm 106.9 e^{j\frac{12\pi}{10}} = \pm 106.9 [-0.809 - j0.59]$
 $= \pm (-86.48 - j63.07)$

For $k=4 \therefore P_4 = \pm 106.9 e^{j\frac{14\pi}{10}} = \pm 106.9 [-0.309 - j0.95]$
 $= \pm (-33.03 - j101.55j)$

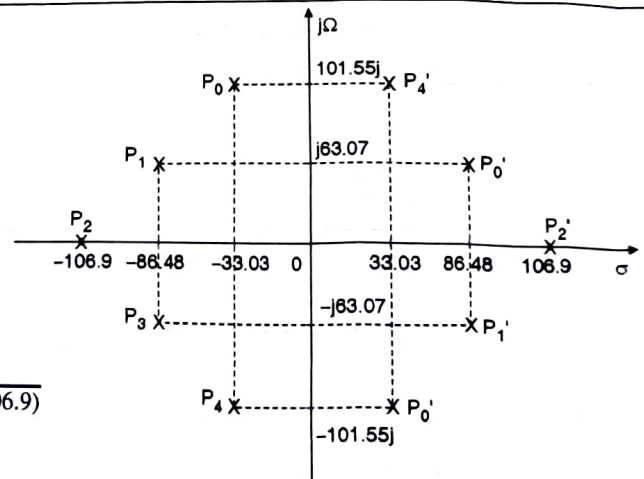


Fig. P. 2.6.10

UEx. 2.6.11 MU - May 16, Dec. 16, 10 Marks

Design a digital Butterworth filter for following specifications using Bilinear transformation technique.

Attenuation in pass band = 1.93dB,

Pass band edge frequency = 0.2π ,

Attenuation in stop band = 13.97dB,

Stop band edge frequency = 0.6π .

✓ **Soln. :**

Given, $A_p = 1.93 \text{ dB}$; $\omega_p = 0.2\pi$; $A_s = 13.97 \text{ dB}$; $\omega_s = 0.6\pi$

Since F_s is not given, consider $T_s = 1 \text{ sec}$

► **Step I :** Using BLT,

$$\Omega_p = \frac{2}{T_s} \tan \frac{\omega_p}{2} = 2 \tan \frac{0.2\pi}{2} = 0.3249$$

$$\text{and } \Omega_s = \frac{2}{T_s} \tan \frac{\omega_s}{2} = 2 \tan \frac{0.6\pi}{2} = 2.75$$

► **Step II :** Order of filter is,

$$N = \frac{1}{2} \frac{\log \left[\frac{10^{0.1A_s} - 1}{10^{0.1A_p} - 1} \right]}{\log \left(\frac{\Omega_s}{\Omega_p} \right)}$$

$$= \frac{1}{2} \cdot \frac{\log \left[\frac{10^{0.1(13.97)} - 1}{10^{0.1(1.93)} - 1} \right]}{\log \left(\frac{2.75}{0.3249} \right)} = \frac{1}{2} \frac{\log \left[\frac{23.94}{0.56} \right]}{\log \left(\frac{2.75}{0.3249} \right)}$$

$$\therefore N = \frac{1}{2} \cdot \frac{1.63}{0.93} = 0.88 \approx 1$$

► **Step III :** Cutoff frequency is,

$$\Omega_c = \frac{\Omega_p}{10^{0.1A_p} - 1} = \frac{0.3249}{10^{0.1(1.93)} - 1} = 0.58$$

► **Step IV :** Calculation of poles

$$P_k = \pm \Omega_c e^{j(N+2k+1)\frac{\pi}{2N}} \dots k = 0 \text{ to } N-1$$

$$\therefore P_k = \pm 0.58 e^{j(1+2k+1)\frac{\pi}{2N}} \dots k = 0$$

$$\therefore P_0 = \pm 0.58 e^{j\pi} = \pm 0.58 (\cos \pi + j \sin \pi)$$

$$\therefore P_0 = \pm 0.58 (-1 + j0) = 0.58, -0.58$$

For stability we will select pole at L.H.S.

$$\therefore P_0 = -0.58$$

► **Step V :** $H(s) = \frac{\Omega_c^N}{s - P_0} = \frac{(0.58)^1}{s + 0.58} = \frac{0.58}{s + 0.58}$

Using BLT, $H(Z) = \frac{0.58}{\frac{1}{2} \left[\frac{Z-1}{Z+1} \right] + 0.58}$

2.7 DESIGNING USING CHEBYSHEV APPROXIMATION

Chebyshev low pass filter design

1. Frequency response

The magnitude squared frequency response of chebyshev filter is given by,

$$|H(j\Omega)|^2 = \frac{1}{1 + \epsilon^2 C_N^2 \left(\frac{\Omega}{\Omega_p} \right)}$$

Here $C_N \left(\frac{\Omega}{\Omega_p} \right)$ = Chebyshev polynomial of order M.

2. Parameter ϵ

It represents ripple parameter in the passband. It is given by,

$$\epsilon = \left[10^{0.1 A_p \text{ dB}} - 1 \right]^{1/2}$$

If A_p is not in dB then ϵ is calculated using the equation,

$$\epsilon = \left[\frac{1}{A_p} - 1 \right]^{1/2}$$

3. At cut-off frequency $\Omega_c = 1$, the magnitude is given by,

$$|H(j\Omega)| = \frac{1}{\sqrt{1 + \epsilon^2}}$$

4. Order of a filter

When magnitude is expressed in dB then order (N) of a filter is obtained by using the equation.

$$|H(j\Omega)| \text{ in dB} = -20 \log_{10} \epsilon - 6(N-1) - 20 \log(\Omega_s)$$

When specification are not in dB then we the equation

$$N = \frac{\cosh^{-1} \left[\frac{1}{\epsilon} \left(\frac{1}{A_p} - 1 \right)^{\frac{1}{2}} \right]}{\cosh^{-1} \left(\frac{\Omega_s}{\Omega_p} \right)}$$

5. Position of poles of chebyshev filter

First calculate regular position of poles of butterworth filter using,

$$\theta_k = \frac{\pi}{2} + \frac{(2k+1)\pi}{2N}, \quad k = 0, 1, 2, \dots, N-1$$

The position of poles of chebyshev filter lie on ellipse at co-ordinates x_k and y_k given by,

Module

2

$$x_k = r \cos \theta_k, \quad k = 0, 1, 2, \dots, N-1$$

$$y_k = R \sin \theta_k, \quad k = 0, 1, 2, \dots, N-1$$

Here r represents minor axis of ellipse and is given by,

$$r = \Omega_p \frac{\beta^2 - 1}{2\beta}$$

and R represents major axis of ellipse and is given by,

$$R = \Omega_p \frac{\beta^2 + 1}{2\beta}$$

Here the parameter β is given by,

$$\beta = \left[\frac{\sqrt{1 + \epsilon^2} + 1}{\epsilon} \right]^{\frac{1}{N}}$$

Then the pole positions are denoted by s_p and,

$$s_p = r \cos \theta_k + jR \sin \theta_k$$

6. System transfer function

The system transfer function of analog filter is given by,

$$H(s) = \frac{k}{(s-s_0)(s-s_1)(s-s_2)\dots}$$

After simplification this equation can be written as,

$$H(s) = \frac{k}{s^N + b_{N-1}s^{N-1} + \dots + b_0}$$

Here b_0 = Constant term in the denominator

Now the value of k is calculated as follows

$$k = \begin{cases} b_0 & \text{for 'N' odd} \\ \frac{b_0}{\sqrt{1 + \epsilon^2}} & \text{for 'N' even} \end{cases}$$

2.7.1 Solved Problems on Chebyshev Approximation

Ex. 2.7.1 : Design a low-pass 1 rad/sec bandwidth chebyshev filter with following characteristics.

- Acceptable pass band ripple of 2 dB.
- Cut-off radian frequency of 1 rad/sec.
- Stop band attenuation of 20 dB or greater beyond 1.3 rad/sec.

Soln. :

Given data : Given cut-off frequency is 1 rad/sec. That means it is normalized low pass chebyshev filter.

Pass band ripple = $A_p = 2$ dB

Stop band attenuation = $A_s = 20$ dB

Stop band frequency = $\Omega_s = 1.3$ rad/sec.

Step I : Calculation of parameter ϵ

We have,

$$\begin{aligned} \epsilon &= \left[10^{0.1 A_p \text{ dB}} - 1 \right]^{\frac{1}{2}} \\ &= \left[10^{0.1 (2)} - 1 \right]^{\frac{1}{2}} = 0.765 \end{aligned}$$

Step II : Calculation of order N of filter

Given stopband attenuation = 20 dB for $\Omega_s = 1.3$ rad/sec

Since it is attenuation we can write,

$$|H(j\Omega)| = -20 \text{ dB for } \Omega = \Omega_s = 1.3 \text{ rad/sec.}$$

The order N of filter is calculated using the equation,

$$\begin{aligned} |H(j\Omega)| \text{ in dB} &= -20 \log_{10} \epsilon - 6(N-1) - 20N \log_{10} \Omega_s \\ \therefore -20 &= -20 \log_{10} (0.765) - 6(N-1) - 20N \log_{10} (1.3) \end{aligned}$$

$$\therefore 6(N-1) + 20N \log_{10} 1.3 = 20 - 20 \log_{10} (0.765)$$

$$6N - 6 + 2.28N = 22.327$$

$$\therefore N = 3.421$$

Since N is order of filter we can write,

$$N \approx 4$$

Step III : Calculation of poles

The pole positions are given by,

$$s_p = r \cos \theta_k + jR \sin \theta_k \quad \dots(1)$$

First we will calculate parameter β

$$\begin{aligned} \beta &= \left[\frac{\sqrt{1 + \epsilon^2} + 1}{\epsilon} \right]^{\frac{1}{N}} \\ &= \left[\frac{\sqrt{1 + (0.765)^2} + 1}{0.765} \right]^{\frac{1}{4}} = 1.31 \quad \dots(2) \end{aligned}$$

Now we will calculate values of r and R .

This is normalized filter. So $\Omega_p = 1$ rad/sec.

We have, minor axis of ellipse = $r = \Omega_p \cdot \frac{\beta^2 - 1}{2\beta}$

$$\therefore r = 1 \cdot \frac{(1.31)^2 - 1}{2(1.31)} = 0.273 \quad \dots(3)$$

The major axis of an ellipse is given by,

$$R = \Omega_p \frac{\beta^2 + 1}{2\beta} = 1 \cdot \frac{(1.31)^2 + 1}{2(1.31)} = 1.04 \quad \dots(4)$$

Now we will calculate values of θ_k as follows :

$$\theta_k = \frac{\pi}{2} + \frac{(2k+1)\pi}{2N}, \quad k = 0, 1, 2, 3.$$

$$\text{For } k = 0 : \quad \theta_0 = \frac{\pi}{2} + \frac{\pi}{8} = \frac{5\pi}{8}$$

$$\text{For } k = 1 : \quad \theta_1 = \frac{\pi}{2} + \frac{3\pi}{8} = \frac{7\pi}{8}$$

$$\text{For } k=2: \quad \theta_2 = \frac{\pi}{2} + \frac{5\pi}{8} = \frac{9\pi}{8}$$

$$\text{For } k=3: \quad \theta_3 = \frac{\pi}{2} + \frac{7\pi}{8} = \frac{11\pi}{8}$$

Using Equation (1), we can calculate pole positions as follows :

$$\text{For } k=0: \quad s_0 = r \cos \theta_0 + jR \sin \theta_0$$

$$\therefore s_0 = 0.273 \cos\left(\frac{5\pi}{8}\right) + j(1.04) \sin\left(\frac{5\pi}{8}\right)$$

$$\therefore s_0 = -0.1 + j0.96$$

$$\text{For } k=1: \quad s_1 = r \cos \theta_1 + jR \sin \theta_1$$

$$\therefore s_1 = 0.273 \cos\left(\frac{7\pi}{8}\right) + j(1.04) \sin\left(\frac{7\pi}{8}\right)$$

$$\therefore s_1 = -0.25 + j0.4$$

$$\text{For } k=2: \quad s_2 = r \cos \theta_2 + jR \sin \theta_2$$

$$\therefore s_2 = 0.273 \cos\left(\frac{9\pi}{8}\right) + j(1.04) \sin\left(\frac{9\pi}{8}\right)$$

$$\therefore s_2 = -0.25 - j0.4$$

$$\text{For } k=3: \quad s_3 = r \cos \theta_3 + jR \sin \theta_3$$

$$\therefore s_3 = 0.273 \cos\left(\frac{11\pi}{8}\right) + j(1.04) \sin\left(\frac{11\pi}{8}\right)$$

$$\therefore s_3 = -0.1 - j0.96$$

Note: Observe that the poles are complex conjugate of each other.

► Step IV : Calculation of system function $H(s)$

The system function $H(s)$ is given by,

$$H(s) = \frac{k}{(s-s_0)(s-s_1)(s-s_2)(s-s_3)}$$

$$\therefore H(s) = \frac{k}{(s+0.1-j0.96)(s+0.25-j0.4)(s+0.25+j0.4)(s+0.1+j0.96)}$$

$$\therefore H(s) = \frac{k}{(s^2+0.2s+0.9316)(s^2+0.5s+0.2225)}$$

$$\therefore H(s) = \frac{k}{s^4+0.7s^3+1.25s^2+0.51s+0.207}$$

Here $N=4$ which is even. Thus

$$k = \frac{b_0}{\sqrt{1+\epsilon^2}}$$

b_0 = Constant term in the denominator = 0.207.

$$\therefore k = \frac{0.207}{\sqrt{1+(0.765)^2}} = 0.16$$

$$\therefore H(s) = \frac{0.16}{s^4+0.7s^3+1.25s^2+0.51s+0.207}$$

This is the required transfer function.

UEx. 2.7.2 MU - Q. 4(a), Dec. 18, 10 Marks

Design a Chebyshev analog filter with maximum pass band attenuation of 2.5 dB at $\Omega_p = 20$ rad/sec and stop band attenuation of 30 dB at $\Omega_s = 50$ rad/sec.

✓ Soln. :

Given data

$$\text{Pass band attenuation} = A_p = 2.5 \text{ dB}$$

$$\text{Pass band frequency} = \Omega_p = 20 \text{ rad/sec}$$

$$\text{Stop band attenuation} = A_s = 30 \text{ dB}$$

$$\text{Stop band frequency} = \Omega_s = 50 \text{ rad/sec}$$

► Step I : Calculation of parameter ϵ

We have,

$$\begin{aligned} \epsilon &= \left[10^{0.1 A_p \text{ dB}} - 1 \right]^{\frac{1}{2}} \\ &= \left[10^{0.1 (2.5)} - 1 \right]^{\frac{1}{2}} = 0.882. \end{aligned}$$

► Step II : Calculation of order N of the filter

Given stop band attenuation = 30 dB for $\Omega_s = 50$ rad/sec.

$$\therefore |H(j\Omega)| = -30 \text{ dB for } \Omega = \Omega_s = 50 \text{ rad/sec.}$$

The order ' N ' of filter is calculated using the equation,

$$|H(j\Omega)| \text{ in dB} = -20 \log_{10} \epsilon - 6(N-1) - 20N \log_{10} \Omega_s$$

$$\therefore -30 = -20 \log_{10} (0.882) - 6(N-1) - 20N \log_{10} (50)$$

$$\therefore -30 = 1.09 - 6(N-1) - 33.98N$$

$$\therefore -30 = 1.09 - 6N + 6 - 33.98N$$

$$\therefore 39.98N = 1.09 + 30 + 6$$

$$\therefore N = 0.95$$

Thus order of filter = $N \approx 1$

► Step III : Calculation of poles

The pole positions are given by,

$$s_p = r \cos \theta_k + jR \sin \theta_k \quad \dots(1)$$

First we will calculate parameter β

$$\begin{aligned} \beta &= \left[\frac{\sqrt{1+\epsilon^2} + 1}{\epsilon} \right]^{\frac{1}{N}} \\ &= \left[\frac{\sqrt{1+(0.882)^2} + 1}{0.882} \right]^1 = 2.64 \quad \dots(2) \end{aligned}$$

We will calculate values of r and R .

We have,

$$\text{Minor axis of ellipse} = r = \Omega_p \cdot \frac{\beta^2 - 1}{2\beta}$$

$$\therefore r = 20 \cdot \frac{(2.64)^2 - 1}{2(2.64)} = 22.6 \quad \dots(3)$$

The major axis of an ellipse is given by,

$$R = \Omega_p \frac{\beta^2 + 1}{2\beta} = 20 \frac{(2.64)^2 + 1}{2(2.64)} = 30.19 \quad \dots(4)$$

Now we will calculate values of θ_k as follows.

$$\theta_k = \frac{\pi}{2} + \frac{(2k+1)\pi}{2N}, \quad k = 0 \text{ to } N-1 \text{ that means for } k=0.$$

$$\therefore \theta_0 = \frac{\pi}{2} + \frac{(0+1)\pi}{2 \times 1} = \pi$$

Using Equation (1) we can write pole positions as,

$$s_0 = r \cos \theta_0 + jR \sin \theta_0 \\ = 22.6 \cos \pi + j(30.19) \sin \pi = -22.6$$

► **Step IV :** Calculation of system function $H(s)$

The system function is given by,

$$H(s) = \frac{k}{(s-s_0)} = \frac{k}{s+22.6}$$

Here for $N=1$ we have,

$$k = b_0 = 22.6$$

$$\therefore H(s) = \frac{22.6}{s+22.6}$$

This is the required transfer function.

Ex. 2.7.3 : Design a digital low pass filter using chebyshev filter design procedure that meets the following specifications : Passband magnitude characteristics that is constant to within 1 dB for frequencies below $\omega = 0.2\pi$ and stop band attenuation of atleast 15 dB for frequencies between $\omega = 0.3\pi$ and π . Use bilinear transformation.

✓ **Soln. :**

In this problem, the specifications of digital filter are given. The given specifications are

$$\text{Passband ripple} = A_p = 1 \text{ dB}$$

$$\text{Passband edge frequency} = \omega_p = 0.2\pi$$

$$\text{Stop band attenuation} = A_s = 15 \text{ dB}$$

$$\text{Stop band edge frequency} = \omega_s = 0.3\pi$$

First we will calculate the values of edge frequencies for the analog filter.

► **Step I :** Calculation of analog filter's edge frequencies

For the bilinear transformation we have,

$$\Omega_p = \frac{2}{T_s} \tan \frac{\omega_p}{2}$$

Note : Sampling time is not given. So we will assume sampling time $T_s = 1 \text{ sec}$.

$$\therefore \Omega_p = 2 \tan \frac{0.2\pi}{2} = 0.6498$$

$$\text{and } \Omega_s = \frac{2}{T_s} \tan \frac{\omega_s}{2}$$

Again $T_s = 1 \text{ sec}$.

$$\therefore \Omega_s = 2 \tan \frac{0.3\pi}{2} = 1.019$$

► **Step II :** Calculation of parameter ϵ

It is given by,

$$\epsilon = \left[10^{0.1 A_p \text{ dB} - 1} \right]^{\frac{1}{2}} \\ = \left[10^{0.1(1) - 1} \right]^{\frac{1}{2}} = 0.5088$$

► **Step III :** Calculation of order (N) of a filter

It is calculated using the equation,

$$|H(j\Omega)| \text{ in dB} = -20 \log_{10} \epsilon - 6(N-1) - 20N \log_{10}(\Omega_s)$$

Given attenuation is 15 dB.

$$\therefore -15 \text{ dB} = -20 \log_{10} 0.5088 - 6(N-1) - 20N \log_{10}(1.019)$$

$$\therefore -15 = 5.87 - 6N + 6 - 0.1634$$

$$\therefore 6.1634N = 26.87$$

$$\therefore N = 4.35$$

Thus order of filter = $N \approx 5$

► **Step IV :** Calculation of pole positions

The pole positions are given by,

$$s_p = r \cos \theta_k + jR \sin \theta_k \quad \dots(1)$$

Now we will calculate values of r , R and θ_k .

$$\text{Minor axis of ellipse} = r = \Omega_p \frac{\beta^2 - 1}{2\beta}$$

$$\epsilon \text{ and } \beta = \left[\frac{\sqrt{1 + \epsilon^2} + 1}{\epsilon} \right]^{\frac{1}{2}}$$

$$\therefore \beta = \left[\frac{\sqrt{1 + (0.5088)^2} + 1}{0.5088} \right]^{\frac{1}{5}} = 1.33$$

$$\therefore r = 0.6498 \cdot \frac{(1.33)^2 - 1}{2 \times 1.33}$$

And Major axis of ellipse = $R = \Omega_p \frac{(1.33)^2 + 1}{2 \times 1.33}$

$$\therefore R = 0.6498 \cdot \frac{(1.33)^2 + 1}{2 \times 1.33}$$

Now we will calculate values of θ_k .

We have,

$$\theta_k = \frac{\pi}{2} + \frac{(2k+1)\pi}{2N}$$

$$\text{For } k=0: \theta_0 = \frac{\pi}{2} + \frac{\pi}{10} = \frac{6\pi}{10}$$

$$\text{For } k=1: \theta_1 = \frac{\pi}{2} + \frac{(2+1)\pi}{10} = \frac{7\pi}{10}$$

$$\text{For } k=2: \theta_2 = \frac{\pi}{2} + \frac{5\pi}{10} = \pi$$

$$\text{For } k=3: \theta_3 = \frac{\pi}{2} + \frac{7\pi}{10} = \frac{9\pi}{10}$$

$$\text{For } k=4: \theta_4 = \frac{\pi}{2} + \frac{9\pi}{10} = \frac{11\pi}{10}$$

Now putting these values in Equation (1)

► **Step V :** Calculation of system transfer function

It is given by,

$$H(s) = \frac{k}{(s-s_0)(s-s_1)(s-s_2)(s-s_3)(s-s_4)}$$

$$\therefore H(s) = \frac{k}{(s+0.186)(s+0.174)(s+0.166)(s+0.158)(s+0.146)}$$

$$\therefore H(s) = \frac{k}{(s+0.186)(s+0.174)(s+0.166)(s+0.158)(s+0.146)}$$

Now $N=5$ thus,

$$k = b_0 = \text{Constant term}$$

$$\therefore k = b_0 = (0.186)(0.174)(0.166)(0.158)(0.146)$$

$$\therefore H(s) = \frac{k}{(s+0.186)(s+0.174)(s+0.166)(s+0.158)(s+0.146)}$$

► **Step VI :** Calculation of transfer function for digital filter

By using bilinear transformation, the transfer function

$$s = \frac{Z-1}{T_s} \left[\frac{Z+1}{Z-1} \right]$$

Here $T_s = \text{Sampling time} = 1 \text{ sec}$

$$\therefore H(Z) = \frac{k}{\left[2 \left(\frac{Z-1}{Z+1} + 0.186 \right) \right] \left[2 \left(\frac{Z-1}{Z+1} + 0.174 \right) \right] \left[2 \left(\frac{Z-1}{Z+1} + 0.166 \right) \right] \left[2 \left(\frac{Z-1}{Z+1} + 0.158 \right) \right] \left[2 \left(\frac{Z-1}{Z+1} + 0.146 \right) \right]}$$

$$\therefore r = 0.6498 \cdot \frac{(1.33)^2 - 1}{2 \times 1.33} = 0.186 \quad \dots(2)$$

And Major axis of ellipse = $R = \Omega_p \cdot \frac{\beta^2 + 1}{2\beta}$

$$\therefore R = 0.6498 \cdot \frac{(1.33)^2 + 1}{2 \times 1.33} = 0.67 \quad \dots(3)$$

Now we will calculate values of θ_k .

We have,

$$\theta_k = \frac{\pi}{2} + \frac{(2k+1)\pi}{2N}, \quad k = 0, 1, 2, \dots, N-1$$

that means $k = 0, 1, 2, 3, 4$

For $k = 0$: $\theta_0 = \frac{\pi}{2} + \frac{\pi}{10} = \frac{6\pi}{10}$

For $k = 1$: $\theta_1 = \frac{\pi}{2} + \frac{(2+1)\pi}{10} = \frac{\pi}{2} + \frac{3\pi}{10} = \frac{8\pi}{10}$

For $k = 2$: $\theta_2 = \frac{\pi}{2} + \frac{5\pi}{10} = \pi$

For $k = 3$: $\theta_3 = \frac{\pi}{2} + \frac{7\pi}{10} = \frac{12\pi}{10}$

For $k = 4$: $\theta_4 = \frac{\pi}{2} + \frac{9\pi}{10} = \frac{14\pi}{10}$

Now putting these values in Equation (1) we get,

For $k = 0$: $s_0 = 0.186 \cos \theta_0 + j 0.67 \sin \theta_0$

$$\therefore s_0 = 0.186 \cos \left(\frac{6\pi}{10} \right) + j 0.67 \sin \left(\frac{6\pi}{10} \right)$$

$$\therefore s_0 = -0.057 + j 0.637$$

For $k = 1$: $s_1 = 0.186 \cos \theta_1 + j 0.67 \sin \theta_1$

$$\therefore s_1 = 0.186 \cos \left(\frac{8\pi}{10} \right) + j 0.67 \sin \left(\frac{8\pi}{10} \right)$$

$$\therefore s_1 = -0.15 + j 0.39$$

For $k = 2$: $s_2 = 0.186 \cos \theta_2 + j 0.67 \sin \theta_2$

$$\therefore s_2 = 0.186 \cos \pi + j 0.67 \sin \pi$$

$$\therefore s_2 = -0.186$$

For $k = 3$: $s_3 = 0.186 \cos \theta_3 + j 0.67 \sin \theta_3$

$$\therefore s_3 = 0.186 \cos \left(\frac{12\pi}{10} \right) + j 0.67 \sin \left(\frac{12\pi}{10} \right)$$

$$\therefore s_3 = -0.15 - j 0.39$$

For $k = 4$: $s_4 = 0.186 \cos \theta_4 + j 0.67 \sin \theta_4$

$$\therefore s_4 = 0.186 \cos \left(\frac{14\pi}{10} \right) + j 0.67 \sin \left(\frac{14\pi}{10} \right)$$

$$\therefore s_4 = -0.057 - j 0.637$$

► **Step V** : Calculation of system transfer function of analog filter

It is given by,

$$H(s) = \frac{k}{(s-s_0)(s-s_1)(s-s_2)(s-s_3)(s-s_4)}$$

$$\therefore H(s) = \frac{k}{(s+0.057-j0.637)(s+0.15-j0.39)(s+0.186)(s+0.15+j0.39)(s+0.057+j0.637)}$$

$$\therefore H(s) = \frac{k}{(s+0.186)(s^2+0.3s+0.1746)(s^2+0.114s+0.409)}$$

Now $N = 5$ thus,

$$k = b_0 = \text{Constant term in the denominator}$$

$$\therefore k = b_0 = (0.186)(0.1746)(0.409) = 0.0135$$

$$\therefore H(s) = \frac{0.0132}{(s+0.186)(s^2+0.3s+0.1746)(s^2+0.114s+0.409)}$$

► **Step VI** : Calculation of transfer function for digital filter

By using bilinear transformation, the transfer function for digital filter $H(Z)$ is obtained from $H(s)$ by putting,

$$s = \frac{2}{T_s} \left[\frac{Z-1}{Z+1} \right]$$

Here T_s = Sampling time = 1 sec. (assumed)

$$\therefore H(Z) = \frac{0.01322}{\left[2 \left(\frac{Z-1}{Z+1} + 0.186 \right) \right] \left\{ \left[2 \left(\frac{Z-1}{Z+1} \right) \right]^2 + 0.3 \left[2 \left(\frac{Z-1}{Z+1} \right) \right] + 0.1746 \right\} \left\{ \left[2 \left(\frac{Z-1}{Z+1} \right) \right]^2 + 0.114 \left[2 \left(\frac{Z-1}{Z+1} \right) \right] + 0.409 \right\}}$$



UEx. 2.7.4 MU - Dec. 2011, 12 Marks

Design a digital Chebyshev filter to satisfy the following constraints :

$$0.707 \leq |H(e^{j\omega})| \leq 1; \quad 0 \leq \omega \leq 0.2\pi$$

$$|H(e^{j\omega})| \leq 0.1; \quad 0.5 \leq \omega < \pi$$

Use bilinear transformation and assume

$T = 1$ second.

✓ **Soln. :**

From the given specifications,

$$A_p = 0.707, \quad A_s = 0.1, \quad \omega_p = 0.2\pi \text{ and } \omega_s = 0.5\pi$$

► **Step I :** Calculation of Ω_p and Ω_s :

Using BLT,

$$\Omega_p = \frac{2}{T_s} \tan \frac{\omega_p}{2} = \frac{2}{1} \cdot \tan \frac{0.2\pi}{2} = 0.6498$$

$$\Omega_s = \frac{2}{T_s} \tan \frac{\omega_s}{2} = \frac{2}{1} \cdot \tan \frac{0.5\pi}{2} = 2$$

► **Step II :** Calculation of ripple parameter ϵ :

The specifications are not given in dB; so use the equation,

$$\epsilon = \left[\frac{1}{A_p^2} - 1 \right]^{\frac{1}{2}} = \left[\frac{1}{(0.707)^2} - 1 \right]^{\frac{1}{2}}$$

► **Step III :** Calculation of order of filter $\epsilon = 1$

When specifications are not in dB then order of filter is calculated using equation,

$$N = \frac{\cos h^{-1} \left[\frac{1}{\epsilon} \left(\frac{1}{A_s^2} - 1 \right)^{\frac{1}{2}} \right]}{\cos h^{-1} \left(\frac{\Omega_s}{\Omega_p} \right)}$$

$$\therefore N = \frac{\cos h^{-1} \left[\frac{1}{1} \left(\frac{1}{(0.1)^2} - 1 \right)^{\frac{1}{2}} \right]}{\cos h^{-1} \left(\frac{2}{0.6498} \right)} = \frac{\cos h^{-1} [9.9498]}{\cos h^{-1} [3.0779]}$$

$$\therefore N = 1.669 \approx 2$$

► **Step IV :** Calculation of poles

The pole positions are given by,

$$s_p = r \cos \theta_k + jR \sin \theta_k$$

The parameter β is given by,

$$\beta = \left[\frac{\sqrt{1+\epsilon^2} + 1}{\epsilon} \right]^{\frac{1}{N}} = \left[\frac{\sqrt{1+1} + 1}{1} \right]^{\frac{1}{2}} = 1.5538$$

Now minor axis of ellipse is,

$$r = \Omega_p \cdot \frac{\beta^2 - 1}{2\beta} = 0.6498 \cdot \frac{(1.5538)^2 - 1}{2 \times 1.5538} = 0.2957$$

The major axis of ellipse is,

$$R = \Omega_p \cdot \frac{\beta^2 + 1}{2\beta} = 0.6498 \cdot \frac{(1.5538)^2 + 1}{2 \times 1.5538} = 0.7139$$

Now we will calculate the values of θ_k as follows

$$\theta_k = \frac{\pi}{2} + \frac{(2k+1)\pi}{2N}, \quad k = 0, 1, \dots, N-1$$

$$\text{For } k=0 \Rightarrow \theta_0 = \frac{\pi}{2} + \frac{\pi}{4} = \frac{3\pi}{4}$$

$$\text{For } k=1 \Rightarrow \theta_1 = \frac{\pi}{2} + \left(\frac{3\pi}{4} \right) = \frac{5\pi}{4}$$

Thus from Equation (1),

For $k=0 \Rightarrow$

$$\begin{aligned} s_0 &= 0.2957 \cos \theta_0 + j(0.7139) \sin \theta_0 \\ &= 0.2957 \cos \left(\frac{3\pi}{4} \right) + j(0.7139) \sin \left(\frac{3\pi}{4} \right) \end{aligned}$$

$$\therefore s_0 = -0.21 + j 0.5$$

For $k=1 \Rightarrow$

$$\begin{aligned} s_1 &= 0.2957 \cos \theta_1 + j(0.7139) \sin \theta_1 \\ &= 0.2957 \cos \left(\frac{5\pi}{4} \right) + j(0.7139) \sin \left(\frac{5\pi}{4} \right) \end{aligned}$$

$$\therefore s_1 = -0.21 + j 0.5$$

► **Step V :** Calculation of transfer function

It is given by,

$$H(s) = \frac{k}{(s-s_0)(s-s_1)} = \frac{k}{(s+0.21-j0.5)(s+0.21+j0.5)}$$

$$H(s) = \frac{k}{s^2 + 0.42s + 0.29} \quad \dots(2)$$

Now order of filter is $N=2$ (even); so k is given by,

$$k = \frac{b_0}{\sqrt{1+\epsilon^2}}$$

Here

b_0 = constant term in denominator of Equation (2)

$$\therefore b_0 = 0.29$$

$$\therefore k = \frac{0.29}{\sqrt{1+1}} = 0.205$$

$$\therefore k = \frac{0.205}{s^2 + 0.42s + 0.29} \quad \dots(3)$$

► **Step VI :** Calculation of $H(z)$

It is obtained by replacing s by $\frac{z-1}{z+1}$ in Equation (3)

$$\begin{aligned}
 H(Z) &= \frac{0.205}{\left[2 \left(\frac{Z-1}{Z+1} \right) \right]^2 + 0.42 \left[2 \left(\frac{Z-1}{Z+1} \right) \right] + 0.29} \\
 &= \frac{0.205}{\frac{4(Z-1)^2}{(Z+1)^2} + 0.84 \left(\frac{Z-1}{Z+1} \right) + 0.29} \\
 &= \frac{0.205 (Z+1)^2}{4(Z-1)^2 + 0.84 (Z-1)(Z+1) + 0.29 (Z+1)^2} \\
 &= \frac{0.205 (Z+1)^2}{4Z^2 - 8Z + 4 + 0.84Z^2 - 0.84 + 0.29Z^2 + 0.58Z + 0.29} \\
 \therefore H(Z) &= \frac{0.205 (Z+1)^2}{5.13Z^2 - 7.42Z + 3.45}
 \end{aligned}$$

Syllabus Topic : Analog and Digital Frequency Transformation with Design Examples

2.8 FREQUENCY TRANSFORMATIONS

Uptill now we have studied the design of low pass filter only. Now if it is asked to design other filter like high pass, band pass or band reject filter then we have to use the frequency transformation.

If the cut-off frequency of LPF is equal to 1 that means if $\Omega_c = 1$ then, it is called as normalized filter. To design the other types of filters; first the system function of normalized LPF is obtained. Then using frequency transformation we can get the system function of the required filter. The following formulae are used for the frequency transformation.

2.8.1 Frequency Transformation in Analog Domain

GQ. Explain frequency transformation in analog domain. (4 Marks)

Let, we have a normalized L.P.F. having cut-off frequency Ω_c .

1. Low pass to low pass

Suppose it is asked to design another LPF with new passband edge frequency Ω_{LP} . Then use the transformation,

$$s \rightarrow \frac{\Omega_c}{\Omega_{LP}} s$$

That means replace 's' by $\frac{\Omega_c}{\Omega_{LP}} s$ in the given equation of $H(s)$.

2. Low pass to high pass

Suppose we have to design HPF with cutoff frequency Ω_{HP}

then use, $s \rightarrow \frac{\Omega_c \Omega_{HP}}{s}$

3. Low pass to bandpass

Suppose we have to design bandpass filter with higher cutoff frequency Ω_u and lower cut-off frequency Ω_l then use the transformation.

$$s \rightarrow \Omega_c \frac{s^2 + \Omega_l \Omega_u}{s(\Omega_u - \Omega_l)}$$

4. Lowpass to bandstop or notch or band reject

Suppose we have to design notch filter with higher cut-off frequency Ω_u and lower cutoff frequency Ω_l then use the transformation,

$$s \rightarrow \Omega_c \frac{s(\Omega_u - \Omega_l)}{s^2 + \Omega_u \Omega_l}$$

The following Table 2.8.1 gives the summary of frequency transformation :

Table 2.8.1

Sr. No.	Conversion type	Transformation
1.	Low pass	$s \rightarrow \frac{\Omega_c}{\Omega_{LP}} s$
2.	High pass	$s \rightarrow \frac{\Omega_c \Omega_{HP}}{s}$
3.	Band pass	$s \rightarrow \Omega_c \times \frac{s^2 + \Omega_l \Omega_u}{s(\Omega_u - \Omega_l)}$
4.	Band stop	$s \rightarrow \Omega_c \times \frac{s(\Omega_u - \Omega_l)}{s^2 + \Omega_u \Omega_l}$

2.8.2 Frequency Transformation in Digital Domain

GQ. Explain frequency transformation in digital domain. (4 Marks)

Similar to analog frequency transformation; a digital low pass filter can be transformed into other digital filters. These relations are follows.

1. Low pass to low pass :

$$Z^{-1} \rightarrow \frac{Z^{-1} \alpha}{1 - \alpha Z^{-1}}$$

$$\text{Here } \alpha = \frac{\sin\left(\frac{\omega'_p - \omega_p}{2}\right)}{\sin\left(\frac{\omega'_p + \omega_p}{2}\right)}$$

ω'_p = Passband frequency of low pass filter which is to be transformed